Java Handbook — Interview Edition

The book you re-read the night before an interview: every chapter of the *Java Handbook* (Java 25 LTS) cut down to the facts, numbers, questions and traps that interviews actually test. Nothing here is new — each line comes from the full book; open the matching chapter there when you need the explanation, the runnable example or its real output.

How to use this edition

- **One section per chapter (1–19)**, always in the same order: **Must-know** (one-line definitions) → **Key facts & numbers** (tables) → **Code to remember** (snippets you may be asked to write) → **In the interview** (likely questions with short answers) → **Gotchas** (mistake → correct idea) → **TL;DR**.
- **Night before:** read the *Top 25 questions* below, then every chapter's **In the interview** and **Gotchas**, then the *Last-minute checklist* at the end.
- **A week before:** read each chapter's **Must-know** and **Key facts**; say each **In the interview** answer out loud in 30 seconds; for anything shaky, go back to the full chapter.
- **Coding rounds:** Chapter 19's pattern table maps problem wording to one of twenty patterns; its trap table lists the Java-specific bugs that pass sample tests and fail hidden ones.
- Versions are marked where they matter: *(16)* means final in Java 16. "Preview" means not final — don't present it as standard.
- Chapter 20 of the full book (cheat sheets and glossary) is folded into the per-chapter tables here.

Top 25 questions across the book

#	Question	One-line answer	Ch
1	JDK vs JRE vs JVM?	JVM runs bytecode + manages memory; JRE = JVM + library; JDK = JRE + tools (`javac`)	1
2	Compiled or interpreted?	Both: `javac` → bytecode; JVM interprets, then JIT-compiles hot code (C1 → C2) using profiles	1, 16
3	Pass-by-value or by reference?	Always by value; for objects the value is a copy of the reference (mutate yes, rebind no)	4
4	Why is `String` immutable?	Thread-safe sharing, cached hash for map keys, pooling, security checks stay valid	7
5	`==` vs `equals`; why is `Integer 128 == 128` false?	`==` compares references; `valueOf` caches only −128…127	2, 7
6	`equals`/`hashCode` contract?	Equal objects must have equal hashes; override both from the same fields, or hash collections lose entries	6
7	How to make a class immutable?	Final class, private final fields, no setters, validate, defensive copies in and out	5
8	Interface vs abstract class?	Interface = contract, many per class, no state; abstract class = shared state/constructors, one parent	6
9	Overloading vs overriding?	Compile-time by argument types vs run-time by object class; statics are hidden	4, 6
10	What are records and sealed types?	Records (16): final immutable data carriers; sealed (17): closed subtype set → exhaustive `switch`	6
11	Checked vs unchecked exceptions?	Checked: catch-or-declare, recoverable external failures; unchecked: programming errors	8
12	How does try-with-resources work?	Closes `AutoCloseable`s in reverse order before catch/finally; close failures become suppressed	8
13	Type erasure and PECS?	Type args removed after checking (no `new T()`, `T[]`, `List<int>`); producer `extends`, consumer `super`	9
14	How does `HashMap` work?	hash → spread → `& (n−1)`; chain → tree (>8, table ≥ 64); resize ×2 at 0.75 load; hash then `equals`	10
15	`ArrayList` vs `LinkedList`; LRU cache?	Array (O(1) get) vs nodes (O(1) ends); LRU = `LinkedHashMap(16, 0.75f, true)` + `removeEldestEntry`	10
16	`Comparable` vs `Comparator`?	Natural order in the class vs external composable orders; never subtract	10
17	`Optional` do's and don'ts?	Return type only; `map`/`flatMap` → `orElse*`/`orElseThrow`; not fields/params/collections	11
18	How do streams execute?	Lazy single-use pipeline; nothing runs until the terminal op; short-circuiting stops early	12
19	`volatile` vs `synchronized` vs atomics?	Visibility only / mutual exclusion + visibility / lock-free single-variable updates	14
20	How is `ConcurrentHashMap` thread-safe?	Lock-free reads, CAS into empty bins, lock a bin's first node; atomic `merge`/`compute`; no nulls	14
21	Why executors; how to configure a pool?	Reuse bounded threads, `Future`s, one shutdown point; bounded queue + rejection policy + named threads	14

#	Question	One-line answer	Ch
22	What is happens-before?	JMM visibility rule: unlock→lock, volatile write→read, start, join, program order, transitive	15
23	Virtual threads — when do they help?	I/O-bound high concurrency (unmount on blocking); not CPU work; don't pool; `Semaphore` for limits	15
24	Deadlock — prevent and detect?	Global lock order, `tryLock` with timeout; thread dump (`jstack`, `jcmd Thread.print`)	15
25	JVM memory areas, GC, leaks?	Heap/stack/metaspace/code cache; reachability-based generational GC, G1 default; leaks = still-referenced objects	16

1. Java and the JVM

What Java is, JDK vs JRE vs JVM, compile → bytecode → run, class loading and the JIT, GC overview, releases and LTS.

Must-know

- **Java** — statically typed, object-oriented language + portable runtime (the JVM). Current LTS: **Java 25** (16 Sept 2025).
- **JVM** — loads, verifies and executes bytecode (interpreting + JIT-compiling) and manages memory. **HotSpot** is the JVM in OpenJDK builds.
- **JRE** — JVM + standard library (enough to *run*). **JDK** — JRE + tools (`javac`, `jshell`, `jar`, `javadoc`, `jlink`). Developers install a JDK.
- **Bytecode** — the portable `.class` format; the *bytecode* is platform-independent, each OS/CPU needs its own JVM.
- **OpenJDK** — the open-source project nearly every JDK (Oracle, Temurin, Corretto, Zulu…) is built from; same code, different support/licensing.
- **Jakarta EE** — enterprise specs (servlets, JPA, CDI) on top of Java SE; formerly Java EE, moved to Eclipse in 2017.
- **Tiered compilation** — interpreter (tier 0) → **C1** (tiers 1–3, fast compile + profiling) → **C2** (tier 4, aggressive optimisation using the profile).
- **Deoptimization** — discard compiled code whose speculative assumption became false; fall back to the interpreter.
- **Stack** — per-thread frames (locals, parameters; only *references* to objects). **Heap** — shared, all objects/arrays, freed by the GC.
- **Metaspace** — class metadata in native memory. **Code cache** — JIT-compiled machine code.
- **Reachability** — GC frees what is unreachable from GC roots (thread stacks, static fields, JNI handles); cycles are collected (no reference counting).
- **Preview feature** — finished but not final; opt in with `--enable-preview` on both `javac` and `java`; may change or be withdrawn (string templates were).
- `--release N` — compile against Java N's API safely (libraries: the oldest Java you support).

Key facts & numbers

Collector	Flag	Good for
G1 (default since Java 9)	`-XX:+UseG1GC`	General servers; region-based, pause goal 200 ms
ZGC	`-XX:+UseZGC`	Large heaps, latency; generational only since JDK 24; pauses typically < 1 ms
Shenandoah	`-XX:+UseShenandoahGC`	Low latency (not in Oracle JDK); generational mode product in 25
Parallel	`-XX:+UseParallelGC`	Throughput / batch
Serial	`-XX:+UseSerialGC`	Tiny heaps, small containers
Epsilon	experimental	Never collects — tests, very short jobs

Java	8	11	17	21	25	26
Class-file major version	52	55	61	65	**69**	70

Approach	Start-up	Peak speed
JIT (HotSpot)	Slower (warm-up)	Very high (live profiles)
AOT cache (JDK 24+) `-XX:AOTCache`	Faster	Same as JIT
GraalVM native image	Very fast	Good, often a bit lower; one binary per OS/CPU

- Release cadence: every **March and September** since Java 10 (2018); **LTS every two years**: 8, 11, 17, 21, 25.
- Milestones: 1996 JDK 1.0 · 2004 Java 5 (generics, enums, autoboxing, annotations, varargs) · 2014 Java 8 (lambdas, streams, `java.time`) · 2017 Java 9 (JPMS, `jshell`) · 2021 Java 17 LTS · 2023 Java 21 LTS (virtual threads, pattern `switch`, record patterns) · 2025 Java 25 LTS.
- Feature stages: **incubator** (`jdk.incubator.*`, e.g. Vector API) → **preview** → **final**; or **withdrawn**.

Code to remember

```
javac -d out src/ch01/hello/HelloWorld.java     # source → bytecode
java -cp out ch01.hello.HelloWorld              # fully qualified name, not the file name
java HelloWorld.java                            # source launch (multi-file since 22)
javap -c -cp out ch01.hello.HelloWorld          # disassemble bytecode
```

In the interview

Q: JDK vs JRE vs JVM? **A:** JVM executes bytecode and manages memory; JRE = JVM + standard library; JDK = JRE + development tools. Install a JDK; ship a `jlink`-trimmed runtime.

Q: Is Java compiled or interpreted? **A:** Both: `javac` compiles to bytecode ahead of time; the JVM interprets it and JIT-compiles hot methods using profiles — so services get faster after warm-up.

Q: How does GC decide what to free? **A:** Reachability from GC roots, not reference counting, so cycles are freed. Most collectors are generational because most objects die young.

Q: When would you pick ZGC? **A:** Large heaps or strict latency targets, where sub-millisecond pauses matter more than a little throughput.

Q: Why is a timing-loop microbenchmark misleading? **A:** It mixes interpreted/C1/C2 phases and the JIT may eliminate "useless" work; use JMH.

Q: What is an LTS? **A:** A release vendors support for years (8, 11, 17, 21, 25); companies standardise on them instead of upgrading every six months.

Q: Name Java 21 features. **A:** Virtual threads, pattern matching for `switch`, record patterns, sequenced collections.

Gotchas

- "Java can't leak memory" → a static, ever-growing `Map` keeps everything reachable = leak.
- `System.gc()` frees memory → it is only a request and usually hurts.
- `finalize()` for cleanup → deprecated for removal (JEP 421, Java 18); use try-with-resources.
- `java Hello.class` → pass the class **name**; packaged classes need the fully qualified name + `-cp`.
- `-Xmx` above the container limit → the JVM reads container limits; let it or size explicitly.
- OpenJDK is "lower quality" than Oracle JDK → same source.
- Compact source file with a `package` line → not allowed; instance `main` was preview until 25.

TL;DR

`javac` → bytecode → JVM loads lazily, verifies, interprets, then JIT-compiles hot code (C1 → C2). Stack = frames, heap = objects, GC frees the unreachable; default G1, ZGC for low latency.

2. Basics: Types, Variables and Operators

The eight primitives, references and `null`, `var`, casting and promotion, operators, overflow, floating point, strings.

Must-know

- **Primitives** — `byte short int long` (8/16/32/64-bit signed), `float double` (IEEE 754), `char` (16-bit unsigned UTF-16 unit), `boolean`. Everything else is a **reference** (or `null`).
- **Locals** have no default — must be assigned before use; **fields** default to `0` / `0.0` / `false` / `'\u0000'` / `null`.
- `final` — assign once; the referenced object can still change.
- `var` (Java 10) — local type inference; the type is still static and fixed. Not for fields, parameters, return types, `var x;`, `var x = null;`, `var a = {1,2};`.
- **Widening** is automatic (but `int/long → float` and `long → double` can lose precision); **narrowing** needs a cast and truncates/wraps.
- **Numeric promotion** — `byte/short/char` become `int` in arithmetic; **compound assignment** (`+=`) hides a cast.
- **Short-circuit** — `&&` / `||` skip the right side; `&` / `|` always evaluate both.
- `==` **on references** compares identity; use `equals` for content.
- **Overflow** wraps silently in two's complement (`MAX_VALUE + 1 == MIN_VALUE`).
- **String** — immutable and pooled; **text blocks** `"""…"""` (Java 15) strip incidental indentation.

Key facts & numbers

Type	Size	Range	Default
`byte`	8 bits	−128 … 127	`0`
`short`	16	−32 768 … 32 767	`0`
`int`	32	≈ ±2.1 billion (-2^{31} … $2^{31}-1$)	`0`
`long`	64	≈ $\pm 9.2 \times 10^{18}$	`0L`
`float`	32	~6–7 significant digits	`0.0f`
`double`	64	~15–16 significant digits	`0.0`
`char`	16	0 … 65 535	`'\u0000'`
`boolean`	unspecified	`true` / `false`	`false`

Rule	Example
Integer `/` truncates toward zero	`-17 / 5 == -3`
`%` takes the sign of the dividend	`-17 % 5 == -2`; use `Math.floorMod` for non-negative
Integer `/ 0`	`ArithmeticException`
Floating `/ 0`	`Infinity`, `-Infinity` or `NaN`
`(int)` cast	truncates toward zero: `(int) -3.99 == -3`
`(byte) 200`	`-56` (low 8 bits)
Octal literal	`010 == 8`
`'a' + 'b'`	`195` (numeric, not string)
`>>` vs `>>>`	arithmetic (copies sign) vs logical (fills 0)
`~x`	`-x - 1`
`1 + 2 + "3" + 4 + 5`	`"3345"` (adds until the first `String`)
`x & 1 == 0`	compile error — write `(x & 1) == 0`

- **Precedence** (high → low): postfix → unary/cast → `* / %` → `+ -` → shifts → relational/`instanceof` → `== !=` → `&` → `^` → `|` → `&&` → `||` → `?:` → assignment/lambda (right-assoc).
- **Overflow tools**: widen early (`(long) a * b`), `Math.addExact/multiplyExact/toIntExact/absExact` (throw), `low + (high - low) / 2`, `BigInteger`.
- **NaN**: test with `Double.isNaN(x)` — never `x == Double.NaN` (NaN ≠ itself). `-0.0 == 0.0` is true but `Double.equals`/`compare` tell them apart.
- **Input**: `Scanner` (easy, slow), `IO.readln` (Java 25), `BufferedReader` (fast, returns `null` at EOF), `System.console()` (passwords; may be `null`).

Code to remember

```
short s = 1;
// s = s + 1;        // compile error: int → short
s += 1;              // OK: implicit cast
int mid = low + (high - low) / 2;          // no overflow
long total = (long) a * b;                 // widen before multiplying
System.out.println(0.1 + 0.2);             // 0.30000000000000004
BigDecimal x = new BigDecimal("0.1");      // not new BigDecimal(0.1)
```

In the interview

Q: Why does `s = s + 1` fail for a `short` but `s += 1` compile? **A:** `s + 1` is promoted to `int` and needs a cast back; compound assignment includes that cast implicitly.

Q: What happens on `int` overflow? **A:** It wraps silently; use `long`, `Math.*Exact` (throws `ArithmeticException`) or `BigInteger`.

Q: Why is `0.1 + 0.2 != 0.3`? **A:** Neither is exactly representable in binary; compare with a tolerance or use `BigDecimal` for money.

Q: Why not `new BigDecimal(0.1)`? **A:** It captures the already-inexact double; use `new BigDecimal("0.1")` or `BigDecimal.valueOf(0.1)`.

Q: Is `long → double` safe? **A:** Magnitude yes, precision no — a double has 53 mantissa bits, a long 64.

Q: Primitive vs reference? **A:** A primitive holds the value; a reference holds a pointer or `null`; `==` compares values vs identity.

Gotchas

- `long big = 9000000000;` → needs `L`; `float f = 1.5;` → needs `f`.
- `7 / 2 == 3` → write `7 / 2.0`; `(double)(7 / 2)` is `3.0` (cast too late).

- `input == "yes"` → `"yes".equals(input)` (null-safe).
- `s.trim();` alone does nothing → strings are immutable; assign the result.
- `"Total: " + a + b` → `"Total: 12"`; parenthesise `(a + b)`.
- `var list = new ArrayList<>()` → `ArrayList<Object>`; `var x = 1` is `int`, not `long`.
- Locals cannot shadow locals: `int x; { int x; }` is a compile error.
- `Scanner.nextInt()` then `nextLine()` → returns the rest of the current line (often empty).

TL;DR

Eight primitives, everything else a reference. Know promotion, truncating `/`, dividend-signed `%`, silent overflow, approximate doubles (`BigDecimal` for money) and `equals` vs `==`.

3. Control Flow

`if`, classic and arrow `switch`, switch expressions, pattern matching (`instanceof`, `switch`, records), loops and their pitfalls.

Must-know

- **Dangling else** — an `else` binds to the nearest unmatched `if`; Java ignores indentation → always use braces.
- **Fall-through** — colon-style `case X:` runs into the next labels until `break`. **Arrow labels** `case X ->` (Java 14) never fall through.
- **Switch expression** — produces a value; inside a block use `yield` (not `return`); must be **exhaustive**.
- **`instanceof` pattern** (Java 16) — `if (o instanceof String s)` tests, casts and binds; `s` is in scope only where the match is certain (`&&`, negated early exit), not with `||`.
- **Pattern `switch`** (Java 21) — type patterns, **`when` guards**, **`case null`**, **dominance** (specific before general, else compile error), exhaustiveness.
- **Sealed + records + pattern switch** = algebraic data types: no `default` needed, adding a subtype breaks every incomplete switch at compile time.
- **Record patterns** (Java 21) — `case Point(int x, var y)` deconstructs, can nest; **unnamed** `_` (Java 22) ignores parts.
- **Loops** — `while` (0+ times), `do-while` (1+ times), `for` (counting), enhanced `for` (every element, no index; compiled to an `Iterator` for `Iterable`s).
- **Labels** — `break outer;` / `continue outer;` target an enclosing loop; Java has no `goto` (reserved word).

Key facts & numbers

Selector type	Allowed since
`char byte short int` + wrappers	1.0 / 5
`enum`	5 (qualified `Day.SAT` labels since 21)
`String` (compared with `equals` after a `hashCode` jump)	7
Any reference type with patterns	21
`long float double boolean`	**not allowed** (primitive patterns still preview, JEP 507)

Situation	Best choice
One condition, two values	ternary
Ranges / unrelated conditions	`if` / `else if`
One variable vs many constants	arrow `switch`
Compute a value from many cases	switch expression
Branch on object type/shape	pattern `switch`
Behaviour per subtype that keeps growing	polymorphism

- A `null` selector throws `NullPointerException` unless there is `case null` (21+).

Code to remember

```
sealed interface Shape permits Circle, Rect {}
record Circle(double r) implements Shape {}
record Rect(double w, double h) implements Shape {}

static double area(Shape s) {
    return switch (s) {                    // exhaustive: no default
        case Circle c -> Math.PI * c.r() * c.r();
        case Rect(double w, double h) when w == h -> w * w;
        case Rect(double w, double h) -> w * h;
    };
}
```

In the interview

Q: What is fall-through? **A:** In colon-style `switch`, execution continues into following labels until `break`; arrow labels never fall through.

Q: What does pattern `switch` give over `instanceof` chains? **A:** Test + cast + bind in one label, `when` guards, explicit `null`, and compile-time exhaustiveness and dominance checks — powerful with sealed types and records.

Q: Why does label order matter? **A:** The first match wins; a label dominated by an earlier one is a compile error.

Q: `while` vs `do-while`? **A:** `while` may run zero times; `do-while` always runs once — "prompt until valid".

Q: Can you switch on a `String`? **A:** Yes (Java 7), compared with `equals`.

Gotchas

- `if (flag = true)` → assignment, always true; write `if (flag)`.
- Unbraced `if` + a second indented line → the second line always runs.
- `default` in an enum switch expression "just in case" → hides the exhaustiveness check for new constants.
- Missing `;` after a switch expression used in an assignment (`};`).
- Removing from an `ArrayList` in a for-each → `ConcurrentModificationException` (or silently wrong when removing the second-to-last) → `Iterator.remove()` / `removeIf`.
- Floating loop counter (`d != 1.0`) → loop on an `int`.
- Assigning the for-each variable → changes only a copy.
- `for (int i = 0; i <= Integer.MAX_VALUE; i++)` → never ends (wraps).
- `list.get(i)` in a loop over a `LinkedList` → O(n²).

TL;DR

Arrow labels and switch expressions (exhaustive, `yield`) replace fall-through; pattern `switch` + sealed records give compiler-checked type dispatch. Watch off-by-one and modifying a collection while iterating.

4. Methods and Arrays

Method anatomy, pass-by-value, overloading, varargs, recursion, arrays and `java.util.Arrays`, `main`.

Must-know

- **Signature** — name + parameter types (not return type, names, modifiers or `throws`).
- **Pass-by-value, always** — primitives copy the value; objects copy the *reference*: mutation through it is shared, reassignment is local. A `swap(a, b)` can't work.
- **Overload resolution** (compile time) — phase 1 exact/widening → phase 2 boxing → phase 3 varargs, then most specific.
- **Varargs** `T... x` — an array parameter; last parameter only, one per method, empty array when no args; `f(int[])` and `f(int...)` are the same signature.
- **Recursion** — base case + smaller call; one stack frame per call; **no tail-call optimisation** → deep recursion throws `StackOverflowError`.
- **Arrays** — fixed size, 0-based, bounds-checked objects with a `length` *field*, default-filled; **covariant** (runtime `ArrayStoreException`).
- **2-D arrays** — arrays of row arrays; can be **jagged**; iterate row-major.
- **`Arrays.asList`** — fixed-size list *backed by* the array: `set` writes through, `add`/`remove` throw.
- **`main`** — classic `public static void main(String[] args)`; `args` is never `null`. **Java 25**: instance/no-arg `main` and compact source files with `IO.println`.

Key facts & numbers

Copy technique	Result
`b = a`	same array, no copy
`a.clone()`	new array, **shallow**
`Arrays.copyOf(a, n)`	truncates or pads with defaults
`Arrays.copyOfRange(a, from, to)`	`to` exclusive
`System.arraycopy(src, sp, dst, dp, len)`	into an existing array; handles overlap

`Arrays` method	Note
`sort` primitives	Dual-Pivot Quicksort, O(n log n) typical, not stable (irrelevant for primitives)
`sort` objects	TimSort — **stable**, O(n log n) worst case
`binarySearch`	sorted input only; absent → `-(insertionPoint) - 1`
`toString` / `deepToString`	readable; `println(a)` prints `[I@…`
`equals` / `deepEquals`	content (array `.equals` is identity)
`fill`, `setAll(i -> …)`	fill with one value / from index
`compare`, `mismatch` (9)	lexicographic / first differing index

Array	`ArrayList`
Fixed size, primitives directly, covariant	Grows, boxes primitives, invariant generics, content `equals`

- Naive Fibonacci $O(\varphi^n) \approx O(1.618^n)$ (quoted as $O(2^n)$); memoized O(n) time and space.

Code to remember

```
static void swap(int[] a, int i, int j) {    // works: mutates the shared array
    int t = a[i]; a[i] = a[j]; a[j] = t;
}
record MinMax(int min, int max) {}           // return two values
static int min(int first, int... rest) { … } // enforce at least one argument
for (int[] row : grid) Arrays.fill(row, 0); // not Arrays.fill(grid, 0)
```

In the interview

Q: Is Java pass-by-reference for objects? **A:** No — strictly pass-by-value; for objects the value is a reference, so a method can mutate the object but reassigning the parameter never affects the caller.

Q: Is `String` special? **A:** No; it only looks different because it is immutable.

Q: How do you return two values? **A:** A small record, e.g. `record MinMax(int min, int max)`.

Q: How is an overloaded call chosen? **A:** At compile time: exact/widening, then boxing, then varargs, then the most specific.

Q: Recursion risks? **A:** Stack frame per call, no TCO → `StackOverflowError`; use memoization for overlapping subproblems or an explicit `ArrayDeque` stack for deep inputs; raising `-Xss` is a last resort.

Gotchas

- Different return type only → not an overload (compile error).
- `list.remove(1)` on `List<Integer>` removes **index** 1 → `list.remove(Integer.valueOf(1))`.
- `f(null)` with `f(String)` and `f(Integer)` → ambiguous; cast.
- `Arrays.asList(intArray)` → `List<int[]>` of one element → `Arrays.stream(a).boxed().toList()`.
- `Arrays.fill(grid, 0)` on `int[][]` → `ArrayStoreException`.
- `grid.length` is the row count; columns are `grid[r].length`.
- Return empty arrays/lists, never `null` (*Effective Java* Item 54).
- `name.toUpperCase();` without assignment does nothing.
- Calling an instance method from `static main` → "non-static method cannot be referenced from a static context".

TL;DR

Everything is passed by value; overloads are chosen at compile time; arrays are fixed-size, covariant, identity-compared objects — use the `Arrays` toolbox to print, compare, sort and copy.

5. OOP: Classes and Objects

Classes and objects, `this`, constructors, packages, `static`, access modifiers, encapsulation, `final`, immutability, initialisation order.

Must-know

- **Class vs object** — blueprint/type vs instance on the heap; variables hold **references**; `null` = no object (helpful NPE messages since Java 14).
- `this` — the current object: reach shadowed fields (`this.size = size`), pass/return yourself (fluent APIs), chain constructors `this(...)`.
- **Constructor** — named like the class, no return type, **not inherited**; the **default no-arg constructor** exists only if you write none; every constructor first calls a `super(...)` (explicit or implicit).
- **Flexible constructor bodies** (Java 25) — validation and computation *before* `this(...)`/`super(...)`, as long as you don't touch `this` (you may assign this class's uninitialised fields).
- **Static** — one per class; no `this`; static methods are **hidden**, not overridden; static init runs **once, lazily**, on first active use (instantiation, static method call, non-constant static field access).

- **Packages** — namespace + folder + access boundary; `java.lang` is auto-imported; wildcard imports do **not** include subpackages; `import module java.base;` (Java 25, JEP 511).
- **Encapsulation** — private state + methods that protect invariants; don't leak mutable internals through getters.
- `final` — variable: assign once; method: no override; class: no subclass. `final` **≠ immutable.**
- **Immutable class** — `final` class, `private final` fields, no setters, validate in the constructor, **defensive copies** of mutable inputs and outputs; "changes" return new objects.
- **Escaping** `this` — registering a listener or starting a thread with `this` in a constructor exposes a half-built object.

Key facts & numbers

Modifier	Class	Package	Subclass (other pkg)	World
`private`	✓			
(none) package-private	✓	✓		
`protected`	✓	✓	✓	
`public`	✓	✓	✓	✓

Way to create objects	Best for
Telescoping constructors	1–3 optional params
Static factory (`List.of` , `Integer.valueOf`)	Names, caching, returning subtypes (EJ Item 1)
Builder	Many optional params (EJ Item 2)
Record	Plain data

Initialisation order: parent statics → child statics (once, source order) → per object: fields get defaults → parent field initialisers/instance blocks → parent constructor body → child field initialisers/blocks → child constructor body.

Code to remember

```
public final class Money {                      // immutable
    private final long cents;
    private final List<String> tags;
    public Money(long cents, List<String> tags) {
        if (cents < 0) throw new IllegalArgumentException("negative");
        this.cents = cents;
        this.tags = List.copyOf(tags);          // defensive copy in
    }
    public Money plus(Money o) { return new Money(cents + o.cents, tags); }
    public List<String> tags() { return tags; }  // already unmodifiable
}
```

In the interview

Q: What does a constructor do and what if you declare none? **A:** Puts a new object in a valid state; with none, Java adds a no-arg default; declare any and it disappears. Chain with `this(...)` so validation lives in one place.

Q: Can a constructor be private? Inherited? **A:** Private yes (singletons, factory-only and utility classes); never inherited.

Q: What is static? **A:** Belongs to the class: one copy, no `this`, hidden not overridden, initialised once on first active use; use for constants, pure utilities, factories — avoid mutable static state.

Q: How do you make a class immutable? **A:** Final class, private final fields, no setters, validate in the constructor, copy mutable things in and out. Gives thread safety, safe sharing/caching, safe map keys.

Q: Why must an immutable class be final? **A:** A subclass could add mutable state or override methods, breaking the guarantee for code holding the parent type.

Q: Is a record deeply immutable? **A:** No — copy mutable components (e.g. `List.copyOf`) in the compact constructor.

Gotchas

- `name = name` in a constructor → assigns the parameter to itself; field stays `null`.
- `public void Pizza() {}` → a method, not a constructor.
- Adding an argument constructor breaks `new Pizza()`.
- Calling overridable methods from a constructor → the child override runs before the child's fields are initialised (EJ Item 19).
- `Collections.unmodifiableList(list)` → a view the owner can still change; `List.copyOf` copies.
- Returning an array field → return `array.clone()`.
- `import java.util.*` + `java.sql.*` then `Date` → ambiguous.
- Static counter `issued++` → race condition across threads (`AtomicInteger`).
- Failure in a static block → `ExceptionInInitializerError`, then `NoClassDefFoundError` on later use.

TL;DR

Constructors initialise (not inherited), `static` is per class, fields private, members as narrow as possible. Immutable = final class + private final fields + no setters + defensive copies; `final` alone is not immutability.

6. OOP: Inheritance, Interfaces and More

Inheritance, overriding, polymorphism, abstract classes, interfaces, sealed types, records, enums, nested classes, `equals` / `hashCode`.

Must-know

- **Inheritance** — `extends`, one parent, root `Object`; is-a. **Prefer composition** (has-a) — the **fragile base class** problem (EJ Items 18–19: design for inheritance or make classes `final`).
- **Overriding** — same signature, covariant return, same-or-wider access, no new/broader checked exceptions, not `final` / `static` / `private`; always `@Override`.
- **Dynamic dispatch** — declared type decides what you may call; runtime type decides what runs. **Static methods and fields** resolve by declared type (**hiding**).
- **Upcast** free; **downcast** checked at runtime (`ClassCastException`) → use `instanceof` patterns, or better, override.
- **Abstract class** — can't instantiate; state + constructors + abstract "holes" (template method, skeletal implementations).
- **Interface** — contract; abstract + constants (`public static final`) + `default` & `static` (8) + `private` (9); no instance state; many per class. Interface static methods are **not inherited**.
- **Default-method conflict** — class wins → more specific interface wins → otherwise override and call `X.super.m()`.
- **Sealed** (17) — `sealed … permits A, B`; subtypes must be `final`, `sealed` or `non-sealed`; enables exhaustive `switch`.
- **Record** (16) — final, immutable data carrier: private final fields, canonical constructor, accessors `x()` (not `getX()`), component-based `equals` / `hashCode` / `toString`. **Compact constructor** for validation/copies. Can implement interfaces; can't extend or add instance fields.
- **Enum** — fixed singleton instances; fields, private constructors, methods, per-constant bodies; `EnumSet` / `EnumMap`.
- **Nested** — static nested (default choice); **inner** holds a hidden outer reference (`outer.new Inner()`, `Outer.this`) → leak risk (EJ Item 24); local/anonymous classes capture effectively final locals.
- **`equals` contract** — reflexive, symmetric, transitive, consistent, `x.equals(null) == false`. **Equal objects must have equal hash codes.**

Key facts & numbers

	Abstract class	Interface
Instance state / constructors	Yes	No
How many per class	One	Many
Method access	Any	public (private helpers since 9)
Use for	Shared state/code	Capability/contract

Need	Reach for
Capability for unrelated classes	Interface
Contract + shared state	Abstract class
Closed set of variants with data	Sealed interface + records
Fixed set of instances	Enum
Immutable data, equal = interchangeable	Record
Mutable object with identity	Class (`final` unless designed for extension)

Enum method	Note
`values()`	new array each call
`valueOf(s)`	exact name or `IllegalArgumentException`
`name()`	persist this
`ordinal()`	never persist (EJ Item 35)

Anonymous class	Lambda
Any interface/abstract class, can have fields, `this` = itself	Functional interface only, no state, `this` = enclosing instance

Code to remember

```
@Override public boolean equals(Object o) {
    if (this == o) return true;
    if (!(o instanceof Book b)) return false;      // class is final
    return year == b.year && Objects.equals(isbn, b.isbn);
}
@Override public int hashCode() { return Objects.hash(isbn, year); }

record Range(int start, int end) {
    Range { if (start > end) throw new IllegalArgumentException(); } // compact
}
```

In the interview

Q: Overloading vs overriding? **A:** Overloading: same name, different parameters, chosen at compile time by argument types. Overriding: same signature in a subclass, chosen at runtime by object type.

Q: Can you override a private or static method? **A:** Private — not inherited, so a same-named method is unrelated. Static — hidden, resolved by declared type.

Q: Abstract class or interface? **A:** Start with an interface for the abstraction clients depend on; add an abstract base only if implementations share real state or code.

Q: Why multiple interface inheritance but not class? **A:** Classes carry state and constructors (diamond ambiguity); interfaces have no instance state and default conflicts have explicit rules.

Q: Why override `equals` and `hashCode` together? **A:** Hash collections bucket by `hashCode` first; equal objects with different hashes are never found.

Q: What if a key's hash changes after insertion? **A:** Lookups probe the wrong bucket; the entry is effectively lost.

Q: Why 31 in hash codes? **A:** Odd prime that spreads well; `31 * h == (h << 5) - h`.

Q: What is a record, and can it be a JPA entity? **A:** Final immutable data carrier with generated members; not with standard JPA (needs no-arg constructor, mutable proxies) — use records as DTOs/projections.

Q: When is sealed a bad fit? **A:** Extension points for other teams/plugins, or variants added often.

Gotchas

- `equals(Book b)` → an overload; `HashSet` never calls it. Use `@Override`.
- Mutable fields in `equals`/`hashCode` of a map key → unfindable after mutation.
- `double` fields compared with `==` → use `Double.compare` (NaN, −0.0).
- `instanceof` vs `getClass()` in `equals` → `instanceof` + `final` class (records do the equivalent); `getClass()` breaks Liskov.
- Subclass constructor without `super(args)` when the parent has no no-arg constructor → compile error.
- Assigning `this.x = x` in a record's compact constructor → compile error; assign the parameter.
- Arrays in records → identity `equals`, breaks value equality.
- Inner (non-static) listener outliving its outer → memory leak.
- Constant interfaces → use a final class or enum (EJ Item 22).

TL;DR

Override with `@Override`, dispatch is on runtime type (statics/fields on declared type). Interfaces for contracts, abstract classes for shared state, sealed + records for closed data, enums for fixed instances; always override `equals` and `hashCode` together.

7. Core APIs: Strings, Numbers, Dates and Randomness

`String` and the pool, `StringBuilder`, formatting, `Math`, wrappers and autoboxing, `BigDecimal`, `java.time`, random numbers.

Must-know

- **`String` is immutable and `final`** — thread-safe, caches its `hashCode` (good map key), keeps security checks valid, enables the **string pool** (on the heap since Java 7, GC'd).
- **Pool** — literals and compile-time constants are pooled; runtime strings aren't → `==` may be true by accident; always `equals`.
- **`StringBuilder`** — mutable, amortised O(1) appends, **not** thread-safe, no value `equals`; `StringBuffer` = synchronized legacy. `+=` in a loop is O(n²); a single `+` expression is fine.
- **Autoboxing** calls `valueOf`, which caches **−128…127** → `==` on `Integer`s is true for 127, false for 128. Unboxing `null` → NPE.
- **`BigDecimal`** — build from `String`, immutable (assign results), `divide` needs scale + `RoundingMode`, compare with **`compareTo`** (`equals` checks scale), **`HALF_EVEN`** (banker's rounding) for money.
- **`java.time`** — immutable and thread-safe. `Instant` = moment (UTC), `LocalDate` = date, `ZonedDateTime` = local time in a region (DST rules), `OffsetDateTime` = fixed offset (APIs, DBs). **`Duration`** (exact seconds) ≠ **`Period`** (calendar units) across DST. Inject a **`Clock`** for testability.
- **DST** — gaps shift forward; overlaps pick the earlier offset.
- **Randomness** — seeded `Random` for reproducible tests, `ThreadLocalRandom` in concurrent code, **`SecureRandom`** for tokens/passwords/session IDs; upper bounds exclusive.

Key facts & numbers

Job	Method	Note
Emptiness	`isEmpty()`, `isBlank()` (11)	blank = whitespace-only too
Whitespace	`trim()` (≤ U+0020) vs `strip()` (11, Unicode)	prefer `strip`
Search	`indexOf`	returns **`-1`** if absent
Replace	`replace` literal vs `replaceAll` **regex**	
Split	`split(regex)`, `split(",", -1)` keeps trailing empties	`split(".")` needs escaping
New in 11	`repeat`, `lines`, `strip*`, `isBlank`	
`compareTo`	<0, 0, >0 (not necessarily ±1)	case-sensitive
`length()`	counts UTF-16 units	not code points

Math	Behaviour
`round`	→ `long` (int for float), ties toward +∞ (`-2.5 → -2`)
`floor` / `ceil` / `rint`	→ `double`; `rint` ties to even
`floorDiv` / `floorMod` (8), `ceilDiv` (18)	round toward −∞/+∞
`clamp` (21)	limit to range
`*Exact` (`powExact` 25)	throw `ArithmeticException` on overflow

	`double`	`long` cents	`BigDecimal`
Exact decimals	No	Yes (2 places)	Yes
Speed	Fastest	Fast	Slow, allocates

- `valueOf` caches −128…127 for `Integer`, `Short`, `Byte`, `Long`; `Character` ≤ 127; both `Boolean`s.
- `int` = 4 bytes; `Integer` ≈ 16 bytes + reference.
- `StringBuilder` default capacity 16 (or 16 + initial length).
- `new String("hi")` creates up to two objects (pooled literal + heap copy).
- Date patterns: `uuuu`/`yyyy` year, **`YYYY` = week-based year** (2025-12-29 → 2026!), `MM` month, `mm` minute, `HH` 0–23, `hh` 1–12, `dd` day, `DD` day-of-year.
- Timing code: `System.nanoTime()` (monotonic), not `Instant.now()`.

Code to remember

```
BigDecimal price = new BigDecimal("19.99");
BigDecimal each  = price.divide(new BigDecimal("3"), 2, RoundingMode.HALF_EVEN);
boolean same = new BigDecimal("1.0").compareTo(new BigDecimal("1.00")) == 0; // true
Instant now = Instant.now(clock);                                  // store UTC
ZonedDateTime local = now.atZone(ZoneId.of("Asia/Kolkata"));       // display
int die = ThreadLocalRandom.current().nextInt(1, 7);               // 1..6
```

In the interview

Q: Why is `String` immutable/final? **A:** Thread safety, cached hash code for map keys, security checks that can't be subverted, and pooling; `final` so no subclass can break it.

Q: `StringBuilder` vs `StringBuffer` vs `+`? **A:** Builder for loops (amortised O(1)), Buffer is synchronized legacy, single `+` expressions are compiler-optimised.

Q: Why is `Integer 127 == 127` true but `128 == 128` false? **A:** `valueOf` caches −128…127; `==` compares references. Use `equals`.

Q: How do you represent money? **A:** `BigDecimal` from strings (or long minor units), explicit scale and `HALF_EVEN`, compare with `compareTo`.

Q: Why is `new BigDecimal("1.0").equals(new BigDecimal("1.00"))` false? **A:** `equals` includes scale; `compareTo` returns 0.

Q: How do you store timestamps? **A:** `Instant` in UTC; convert to `ZonedDateTime` with the user's `ZoneId` only for display; `LocalDate` for dates.

Q: `ZonedDateTime` vs `OffsetDateTime`? **A:** Zoned knows the region's DST rules; Offset is a fixed offset — better for serialising a moment.

Q: Why was `SimpleDateFormat` a problem? **A:** Mutable, not thread-safe; a shared static instance corrupted results. `DateTimeFormatter` is immutable.

Gotchas

- `"TITLE".toLowerCase()` in Turkish locale → `toLowerCase(Locale.ROOT)` for identifiers.
- `sb1.equals(sb2)` → identity; `new StringBuilder('x')` → capacity 120, empty.
- `Long.equals(Integer)` always false; `map.get(42)` on `Map<Long,…>` → `null` (use `42L`).
- Mixed `Integer`/`int` ternary with a `null` → NPE on unboxing.
- `synchronized (someInteger)` → shared cached instances.
- `Boolean.parseBoolean("YES")` → `false`.
- `2.0` vs `2.00` `BigDecimal` → different `HashSet` keys, same `TreeSet` key.
- Unbounded `divide` (1/3) → `ArithmeticException`.
- `LocalDateTime` for event timestamps → no zone, not a moment.
- `"EST"`/`"IST"` abbreviations → ambiguous; use region IDs.
- `Math.abs(r.nextInt()) % n` → biased and can be negative (`MIN_VALUE`); use `nextInt(n)`.

TL;DR

Strings immutable (`equals`, `StringBuilder` in loops), wrappers cache −128…127 (`equals`, beware null unboxing), `BigDecimal` from strings with `compareTo`/`HALF_EVEN`, `Instant` to store and `ZonedDateTime` to display, `SecureRandom` for secrets.

8. Exceptions

The `Throwable` hierarchy, checked vs unchecked, `try`/`catch`/`finally`, try-with-resources, custom exceptions, chaining, best practices.

Must-know

- **Hierarchy** — `Throwable` → `Error` (JVM problems; don't catch) and `Exception`. **Unchecked** = `RuntimeException` + `Error` (and subclasses); every other `Exception` is **checked**.
- **Handle or declare** — checked exceptions must be caught or listed in `throws`; overrides may not throw new/broader checked exceptions.
- **Propagation** — `throw` stops the block; the stack unwinds to the first matching `catch`; uncaught → the thread dies with a stack trace.
- **`finally`** always runs (except `System.exit`, JVM crash, infinite loop); a return value is already evaluated before `finally` runs.
- **Catch rules** — first match wins; subclass before superclass; multi-catch types must be unrelated and the variable is implicitly `final`; unnamed `catch (X _)` (22).
- **try-with-resources** — any `AutoCloseable`; closed in **reverse order**, **before** `catch`/`finally`; if body and `close` both throw, the body's exception wins and close failures become **suppressed** (`getSuppressed()`). `try (existingVar)` since Java 9.
- **Chaining** — always pass the cause: `new ServiceException("msg", e)`; translate to your layer's abstraction; `UncheckedIOException` inside lambdas.
- **Checked vs unchecked** (EJ 70–71) — checked for recoverable external conditions, runtime for programming errors; avoid checked when the caller can't act. `java.util.function` interfaces declare no checked exceptions.

- `Optional` for normal absence; exceptions only for exceptional conditions (EJ 69).

Key facts & numbers

Exception	Kind	Cause
`NullPointerException`	unchecked	dereferencing/unboxing `null`
`IllegalArgumentException` / `NumberFormatException`	unchecked	bad argument / `parseInt("abc")`
`IllegalStateException`	unchecked	call at the wrong time
`ArithmeticException`	unchecked	int `/ 0`, `*Exact` overflow, non-terminating `BigDecimal`
`UnsupportedOperationException`	unchecked	`List.of(...).add(x)`
`ConcurrentModificationException`	unchecked	modify while iterating
`NoSuchElementException`	unchecked	`next()` past end, empty `Optional.get()`
`IOException`, `SQLException`, `InterruptedException`	checked	I/O, JDBC, interrupted blocking
`ExecutionException`, `TimeoutException`	checked	`Future` failure / wait too long
`ExceptionInInitializerError`, `StackOverflowError`, `OutOfMemoryError`	error	static init, runaway recursion, heap exhausted

Situation	Throw
Invalid argument value	`IllegalArgumentException`
Null argument	`Objects.requireNonNull(x, "x")` → NPE
Bad index	`Objects.checkIndex(i, size)` (9)
Wrong object state	`IllegalStateException`
Unsupported by this implementation	`UnsupportedOperationException`

`finally` situation	Result
`return` in `try`, `finally` changes the variable	primitive result unchanged
`return` in `finally`	overrides the return **and discards the exception**
`finally` throws	original exception lost

Code to remember

```
try (var in = Files.newBufferedReader(src);
     var out = Files.newBufferedWriter(dst)) {       // closed out, then in
    in.transferTo(out);
} catch (IOException e) {
    throw new UncheckedIOException("copy " + src + " failed", e);  // keep the cause
}
catch (InterruptedException e) { Thread.currentThread().interrupt(); }  // restore flag
```

In the interview

Q: Checked vs unchecked? **A:** Checked (compiler-enforced catch-or-declare) for recoverable external failures like `IOException`; unchecked `RuntimeException`s for programming errors. `Error` is unchecked and not meant to be caught.

Q: Should you catch `Throwable`? **A:** Only at the very top of a thread/request to log and fail cleanly — it also catches `OutOfMemoryError`.

Q: How does try-with-resources handle two failures? **A:** The body's exception propagates; close failures are attached as suppressed — nothing lost.

Q: `AutoCloseable` vs `Closeable`? **A:** `Closeable` (java.io) extends `AutoCloseable`, throws `IOException`, and its `close` should be idempotent.

Q: `final` vs `finally` vs `finalize`? **A:** Modifier; always-run block; deprecated GC hook (JEP 421, Java 18) — use try-with-resources or `Cleaner`.

Q: What's wrong with `catch (Exception e) {}`? **A:** It hides every failure, including bugs, leaving unknown state.

Q: Why restore the interrupt flag? **A:** Catching `InterruptedException` clears it; re-setting lets callers see cancellation was requested.

Q: Your exception-handling philosophy? **A:** Validate early, standard unchecked exceptions for contract violations, checked only when recovery is realistic, translate with cause at boundaries, never swallow, log once, `Optional` for expected absence.

Gotchas

- `throw new X("failed: " + e.getMessage())` → loses the type and trace; pass `e`.
- Log-and-rethrow at every layer → duplicate logs; log once where handled.
- Returning a `Files.lines` stream out of a `try` → already closed.
- `catch (RuntimeException) {} catch (ClassCastException)` → compile error (already caught).
- Catching a checked exception the body can't throw → compile error (except `Exception`/`Throwable`).
- Exceptions for control flow → slow (stack capture) and hides bugs.
- Mutate before validating → violates failure atomicity (EJ 76).
- Reading traces: top line → first frame in *your* code → last "Caused by:".

TL;DR

Unchecked for bugs, checked for recoverable failures; try-with-resources for anything closeable (suppressed, not lost); chain causes; never swallow or `return` from `finally`.

9. Generics

Generic classes and methods, inference, bounds, invariance and wildcards (PECS), type erasure and its restrictions.

Must-know

- **Purpose** — compile-time type safety, no casts. **Raw types** (`List`) are legacy; use `List<?>` if you don't care.
- **Diamond** `new Box<>()` (Java 7; with anonymous classes since 9). **Generic method**: `<T>` goes after modifiers, before the return type — `public static <T> T pick(...)`.
- **Bounds** — `<T extends Number>`; multiple `<T extends A & I1 & I2>` (class first); recursive `<T extends Comparable<? super T>>`.
- **Invariance** — `List<Integer>` is **not** a `List<Number>` (else you could add a `Double`). Arrays are covariant and reified → runtime `ArrayStoreException`.
- **PECS** — Producer `extends`, Consumer `super`; both → exact `T`; don't care → `?`. Wildcards on parameters, **never in return types**.
- **Type erasure** — type arguments are checked, then removed; `T` → its bound or `Object`; casts inserted; **bridge methods** keep overriding working.
- **Heap pollution** — a variable of a parameterized type refers to an object of another type (unchecked ops); `@SafeVarargs` only for non-escaping varargs arrays.
- **Super type tokens** — `new TypeReference<List<User>>() {}` keeps the generic superclass in the class file (Jackson-style).

Key facts & numbers

Wildcard	Read as	Can add	Accepts
`List<? extends Number>`	`Number`	nothing (only `null`)	`List<Integer>`, `List<Double>` …
`List<? super Integer>`	`Object` only	`Integer`	`List<Integer>`, `List<Number>`, `List<Object>`
`List<?>`	`Object`	nothing (only `null`)	any list
`List<Number>`	`Number`	any `Number`	only `List<Number>`

You can't	Workaround
`List<int>`	`List<Integer>`, `IntStream`, `int[]`
`new T()`	`Supplier<T>` or `Class<T>` token
`new T[n]`	`List<T>`; `Array.newInstance`; internal `(T[]) new Object[n]`
`static T field`	static generic methods declare their own `<T>`
`o instanceof List<String>` (from `Object`)	`instanceof List<?>`
Overload `f(List<String>)` / `f(List<Integer>)`	different names (same erasure)
Generic `Throwable` subclass	concrete field for extra data

Java (erasure)	C# (reified)
Runtime doesn't know `List<String>`; no `new T()`; primitives need boxing	Runtime knows; `new T()` and `List<int>` allowed

- Type-parameter letters: `T` type, `E` element, `K`/`V` key/value, `R` result.
- EJ Items 26–33: no raw types, eliminate unchecked warnings, lists over arrays, favour generic types/methods, bounded wildcards, careful generics+varargs, typesafe heterogeneous containers (`<T> T get(Class<T>)`).

Code to remember

```
static <T extends Comparable<? super T>> T max(List<? extends T> xs) {
    T best = xs.get(0);
    for (T x : xs) if (x.compareTo(best) > 0) best = x;
    return best;
}
static <T> void copy(List<? super T> dest, List<? extends T> src) {
    for (T x : src) dest.add(x);                    // PECS
}
```

In the interview

Q: Why can't `List<Integer>` be passed as `List<Number>`? **A:** Generics are invariant — otherwise you could add a `Double` into a list of integers. Use `List<? extends Number>`.

Q: Explain PECS. **A:** A parameter you read from (producer) is `? extends T`; one you write into (consumer) is `? super T`; `Collections.copy(List<? super T> dest, List<? extends T> src)`.

Q: Why can't you add to `List<? extends Number>`? **A:** It might be a `List<Double>`; only `null` is safe.

Q: What is type erasure and what does it forbid? **A:** Type arguments are removed after compile-time checks, so no `new T()`, generic arrays, primitive type arguments, overloads differing only by type argument.

Q: Bridge method? **A:** A synthetic method so overriding still works after erasure changes a signature.

Q: `<T extends Comparable<T>>` vs `<T extends Comparable<? super T>>`? **A:** The second accepts types that inherit `compareTo` from a supertype (`Apple extends Fruit implements Comparable<Fruit>`).

Q: Arrays vs generics? **A:** Arrays covariant + reified (runtime checks); generics invariant + erased (compile-time checks).

Gotchas

- `new ArrayList()` (forgot `<>`) → raw type.
- `<T>` on a method of `Box<T>` → shadows the class's `T`.
- `<T> void print(List<T>)` with `T` used once → `List<?>`.
- `List<Object>` parameter → callers with `List<String>` can't pass it.
- `@SuppressWarnings("unchecked")` at class level → suppress narrowly, with a comment.
- Generics give **no runtime checks** → `Collections.checkedList` to find polluting code.

TL;DR

Invariant generics + PECS wildcards on parameters; erasure explains every restriction (no `new T()`, no `T[]`, no primitives); never raw types.

10. Collections

The framework hierarchy, lists, sets, maps, `HashMap` internals, queues, fail-fast iterators, ordering, immutable and sequenced collections, big-O.

Must-know

- **Hierarchy** — `Iterable` → `Collection` → `List` / `Set` / `Queue` → `Deque`; `Map` **is separate** (views: `keySet`, `values`, `entrySet`). Program to the interface.
- `ArrayList` — growable array (capacity 10 on first add, then ×1.5): O(1) `get`, amortised O(1) append, O(n) front/middle edits. **The default list.**
- `LinkedList` — doubly linked nodes: O(1) at ends, O(n) `get(i)`, ~24 B per node, poor cache locality. Rarely the best choice.
- **Sets** — `HashSet` (backed by `HashMap`, O(1)), `LinkedHashSet` (insertion order), `TreeSet` (red-black tree, sorted, O(log n), `floor`/`ceiling`), `EnumSet` (bit vector).
- **Maps** — `HashMap` default, `LinkedHashMap` (insertion or access order → **LRU** via `removeEldestEntry`), `TreeMap` (sorted keys, `floorKey`, `ceilingEntry`, `headMap`, `subMap`).
- `ArrayDeque` — circular array for both **queue and stack**; rejects `null`. Prefer it to `Stack` (extends synchronized `Vector`) and `LinkedList`.
- `PriorityQueue` — binary heap: O(log n) `offer`/`poll`, O(1) `peek`, **iteration is not sorted**.
- **Fail-fast iterators** — structural change during iteration → `ConcurrentModificationException` (best effort; a bug detector, not a lock). Use `removeIf` / `Iterator.remove`.
- `Comparable` = one natural order (`compareTo`); `Comparator` = any order, composed with `comparing`/`thenComparing`/`reversed`/`nullsLast`. Sorted collections treat `compare == 0` as equal → keep consistent with `equals`. Object sorts are **stable** (TimSort).
- **Immutable factories** — `List/Set/Map.of` (9), `copyOf` (10): reject `null`, shallowly immutable. `Collections.unmodifiableX` is only a **view**. `Stream.toList()` (16) unmodifiable but allows nulls.
- **Sequenced collections** (21) — `getFirst/getLast/addFirst/addLast/removeFirst/removeLast/reversed()` on lists, deques, `LinkedHashSet`, sorted sets; maps get `firstEntry/lastEntry/pollFirstEntry/putFirst`. `reversed()` is a view.

Key facts & numbers

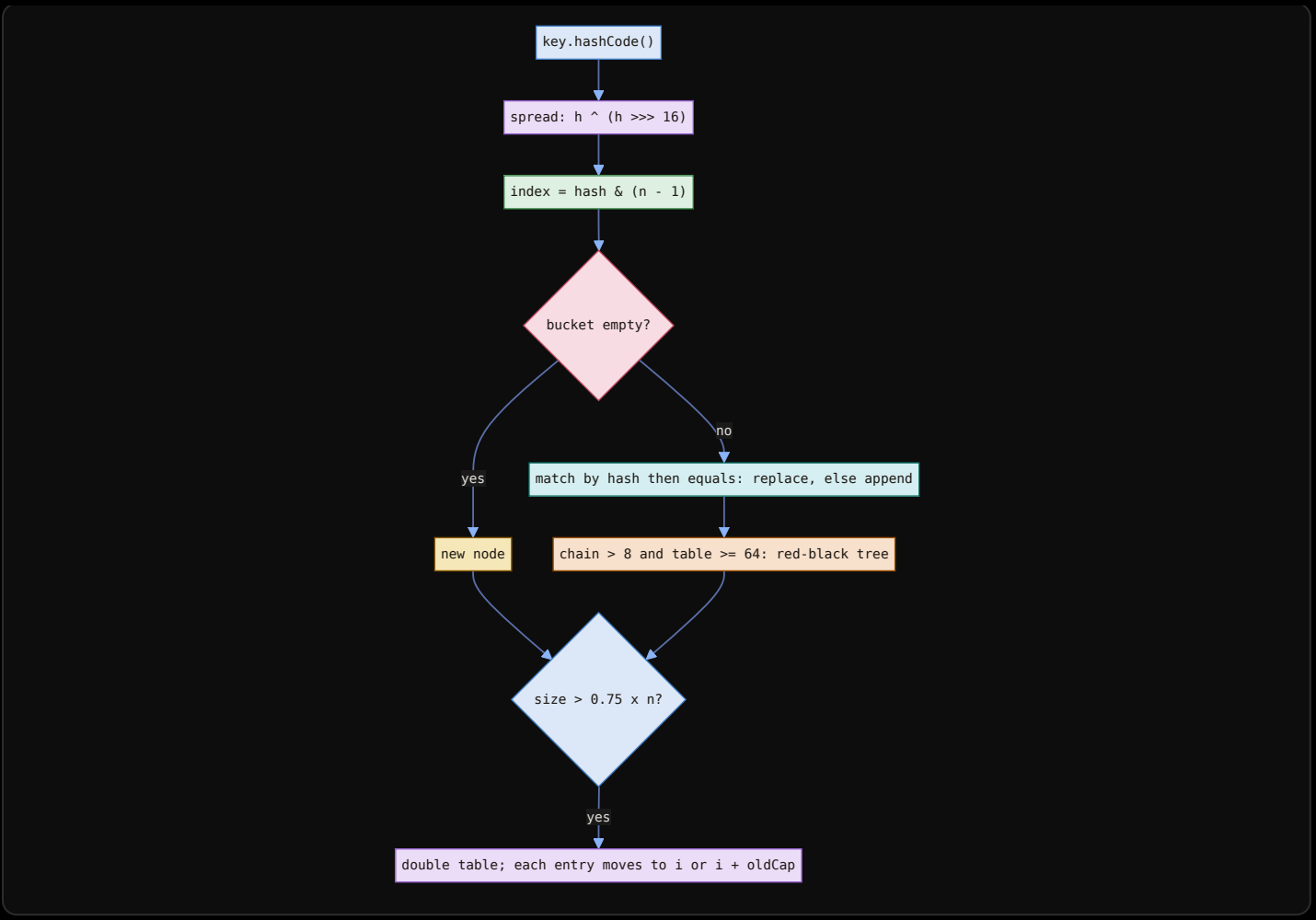
Collection	add	front add/remove	get(i)	contains/get(key)	Order	Nulls
`ArrayList`	O(1)*	O(n)	O(1)	O(n)	index	yes
`LinkedList`	O(1)	O(1)	O(n)	O(n)	index	yes
`ArrayDeque`	O(1)*	O(1)*	—	O(n)	head→tail	**no**
`PriorityQueue`	O(log n)	poll O(log n)	—	O(n)	heap	no
`HashSet` / `HashMap`	O(1)†	—	—	O(1)†	none	one null key
`LinkedHash*`	O(1)†	—	—	O(1)†	insertion/access	one null key
`TreeSet` / `TreeMap`	O(log n)	—	—	O(log n)	sorted	no (natural)
`CopyOnWriteArrayList`	O(n)	O(n)	O(1)	O(n)	index	yes

* amortised · † average

`HashMap` constant	Value
Default capacity (allocated lazily on first `put`)	16
Load factor → resize (double) when size > cap × 0.75	0.75
Treeify bucket at chain length	> 8 (if table ≥ 64, else resize)
Untreeify at	6
Max capacity	2^{30}

Queue op	Throws	Special value
insert	`add`	`offer` → `false`
remove head	`remove`	`poll` → `null`
examine	`element`	`peek` → `null`

Factory	Nulls	Duplicates
`List.of`	NPE	allowed
`Set.of`	NPE	**IAE**; iteration order randomised per run
`Map.of` (≤ 10 pairs) / `Map.ofEntries`	NPE	duplicate key → IAE
`Arrays.asList`	allowed	fixed-size, `set` writes through



Code to remember

```
Map<String, Integer> freq = new HashMap<>();
for (String w : words) freq.merge(w, 1, Integer::sum);          // counting
Map<Integer, List<String>> byLen = new HashMap<>();
for (String w : words) byLen.computeIfAbsent(w.length(), k -> new ArrayList<>()).add(w);
list.removeIf(x -> x < 0);                                      // safe removal
people.sort(Comparator.comparing(Person::city)
        .thenComparing(Person::age, Comparator.reverseOrder()));

class Lru<K, V> extends LinkedHashMap<K, V> {
    private final int cap;
    Lru(int cap) { super(16, 0.75f, true); this.cap = cap; }      // access order
    @Override protected boolean removeEldestEntry(Map.Entry<K, V> e) { return size() > cap; }
}
```

In the interview

Q: `ArrayList` vs `LinkedList`? **A:** ArrayList: O(1) random access, amortised O(1) append, O(n) middle edits. LinkedList: O(1) ends, O(n) indexing, more memory, poor locality. Default to ArrayList; ArrayDeque for queues/stacks.

Q: How does `HashMap` work? **A:** hashCode → spread high bits → `& (n−1)` picks a bucket; collisions chain, and a bucket over 8 entries becomes a red-black tree (table ≥ 64), so worst case O(log n). Past 0.75 load the table doubles; entries move to `i` or `i + oldCap`. Lookup compares hash then `equals`.

Q: Why is capacity a power of two? **A:** `hash & (n−1)` replaces modulo, and resizing splits each bucket into two predictable positions.

Q: Null keys? **A:** `HashMap` allows one (hash 0); `Hashtable` and `ConcurrentHashMap` reject nulls; `TreeMap` rejects with natural ordering.

Q: Same hashCode, not equal? **A:** Same bucket, distinguished by `equals` — correct, just slower.

Q: HashSet vs TreeSet? **A:** O(1) unordered vs O(log n) sorted with range queries; TreeSet uses compare-to-zero for equality.

Q: Why `ArrayDeque` over `Stack`? **A:** `Stack` extends synchronized `Vector` (locking overhead, index methods); `ArrayDeque` is a lean circular array.

Q: Build an LRU cache? **A:** `LinkedHashMap` with `accessOrder = true` + `removeEldestEntry`.

Q: `List.of` vs `Arrays.asList`? **A:** `List.of` fully unmodifiable, no nulls; `Arrays.asList` fixed-size writable view, nulls allowed.

Q: Is `List.of` thread-safe? **A:** Yes for reading (immutable); the elements may not be.

Gotchas

- `equals` without `hashCode` → duplicates in `HashSet`, lost `HashMap` keys.
- Mutating a key inside a `HashMap` / `TreeSet` → unfindable.
- `map.get(k) != null` as presence test → use `containsKey` when values may be `null`.
- `map.get(k) + 1` for absent `k` → NPE (unboxing); use `merge` / `getOrDefault`.
- Iterating `keySet()` then `get` → two lookups; iterate `entrySet()`.
- `computeIfAbsent` whose function modifies the same `HashMap` → CME (Java 9+).
- Subtraction comparators overflow → `Integer.compare` / `comparingInt`.
- Comparator returning 0 for different elements → `TreeSet` silently drops them.
- Printing a `PriorityQueue` → not sorted; `poll()` in a loop.
- `push` adds at the **front** of a deque.
- Forward index loop with `remove(i)` → skips the shifted element.
- Relying on `HashMap` / `HashSet` iteration order → can change on resize or JDK version.
- `subList` then structurally modifying the parent → CME.

TL;DR

`ArrayList`, `HashMap`, `HashSet`, `ArrayDeque` by default; `Tree*` for sorted/range, `LinkedHash*` for order/LRU, `PriorityQueue` for "next smallest". Know HashMap's 16 / 0.75 / 8 / 64 and keep `equals` / `hashCode` consistent and keys immutable.

11. Lambdas and Functional Programming

Functional interfaces, lambda syntax, `java.util.function`, method references, effectively final capture, composition, `Optional`.

Must-know

- **Functional interface** — exactly one abstract method (`@FunctionalInterface` documents and checks it); a lambda's type comes from its **target type**.
- **Lambda syntax** — `x -> expr`, `(a, b) -> { …; return r; }`; types all-or-none; `(var a, var b)` (11) for annotations; `_` for unused parameters (22).
- **Lambda vs anonymous class** — lambda: functional interfaces only, no state, **`this` = enclosing instance**, compiled via `invokedynamic` + private method. Anonymous class: any interface/class, fields, `this` = itself, separate class file.
- **Effectively final capture** — captured locals are copied, so they must never be reassigned; fields are unrestricted; the captured *object* can still be mutated.
- **Method references** — `Class::staticMethod`, `obj::method` (bound), `Class::instanceMethod` (unbound: first argument is the receiver), `Class::new`.
- **Composition** — `f.andThen(g)` = g(f(x)); `f.compose(g)` = f(g(x)); `Predicate.and/or/negate`, `Predicate.not` (11).
- **`Optional`** — return type for "maybe a value"; chain `map` / `flatMap` / `filter`, finish with `orElse` / `orElseGet` / `orElseThrow` / `ifPresentOrElse`. Not for fields, parameters or collections; never return `null` from it; not `Serializable`.
- **Pure function** — output depends only on input, no side effects → easy to test, cache, parallelise.

Key facts & numbers

Interface	Shape	Method
`Supplier<T>`	`() → T`	`get`
`Consumer<T>` / `BiConsumer<T,U>`	`T → void`	`accept`
`Predicate<T>` / `BiPredicate`	`T → boolean`	`test`
`Function<T,R>` / `BiFunction<T,U,R>`	`T → R`	`apply`
`UnaryOperator<T>` / `BinaryOperator<T>`	`T → T` / `(T,T) → T`	`apply`
Primitive: `IntPredicate`, `ToIntFunction<T>`, `IntSupplier.getAsInt` …	avoid boxing	

Optional	Use
`of(v)`	NPE if `v` null (null is a bug)
`ofNullable(v)`	empty if null
`map` vs `flatMap`	use `flatMap` when the function already returns `Optional`
`orElse(x)` vs `orElseGet(s)`	`orElse` evaluates `x` **eagerly**; `orElseGet` lazily
`orElseThrow()` (10)	same as `get()` with an honest name
`OptionalInt/Long/Double`	for primitives

Code to remember

```
Function<String, Integer> len = String::length;          // unbound
Consumer<Object> print = System.out::println;            // bound
Supplier<List<String>> mk = ArrayList::new;              // constructor
Predicate<String> nonBlank = Predicate.not(String::isBlank);
String city = findUser(id)
        .map(User::address)
        .flatMap(Address::city)          // city() returns Optional<String>
        .orElse("unknown");
```

In the interview

Q: What is a functional interface? **A:** An interface with exactly one abstract method (defaults/statics allowed); lambdas and method references implement it.

Q: Why must captured variables be effectively final? **A:** The lambda gets a copy of the value; allowing reassignment would make the copy and the variable diverge (and invite races).

Q: `this` in a lambda vs anonymous class? **A:** Lambda: the enclosing instance. Anonymous class: the anonymous object.

Q: How should `Optional` be used? **A:** As a return type for normal absence; chain `map` / `flatMap` / `filter` and end with `orElse*` / `orElseThrow`; not for fields, parameters or collections.

Q: `orElse` vs `orElseGet`? **A:** `orElse` always evaluates its argument; use `orElseGet` when the default is expensive.

Q: Is Java functional? **A:** No — object-oriented with functional features; functions are objects implementing interfaces.

Gotchas

- `x -> { x * 2 }` → block needs `return x * 2;`.
- Capturing a classic `for` index `i` → reassigned by `i++`; copy it or use `IntStream.range`.
- Accumulating via `int[] c = {0}` or `AtomicInteger` → shared mutable state; compute with `sum` / `count` / `collect`.
- Lambda parameter named like an existing local → "already defined".

- `if (opt.isPresent()) opt.get()` → verbose null check; `Optional<List<T>>` → return an empty list.
- Checked exceptions can't escape `Function` / `Consumer` → wrap deliberately (e.g. `UncheckedIOException`) or use a loop.

TL;DR

Lambda = implementation of a one-method interface, typed by context; capture only effectively final locals; use method references when clearer; `Optional` is for return values only.

12. Streams

Lazy pipelines, intermediate and terminal operations, `reduce`, collectors, primitive streams, parallel streams, gatherers.

Must-know

- **Stream** — a one-shot, lazy pipeline: **source → intermediate ops → one terminal op**. Doesn't store data, never modifies its source, can be infinite, traversed **once** (reuse → `IllegalStateException`).
- **Laziness** — nothing runs without a terminal op; elements flow **one at a time** through fused stages; **short-circuit** ops (`limit`, `findFirst`, `anyMatch`) stop early; **stateful** ops (`sorted`, `distinct`) buffer. Filter early, sort late.
- **Intermediate** — `filter`, `map` (1→1), `flatMap` (1→many), `mapMulti` (16), `distinct`, `sorted`, `limit`, `skip`, `takeWhile` / `dropWhile` (9), `peek` (debug only).
- **Terminal** — `toList()` (16, unmodifiable, nulls ok), `collect`, `count`, `min` / `max`, `findFirst` / `findAny`, `anyMatch` / `allMatch` / `noneMatch`, `reduce`, `forEach`.
- `reduce` — needs a true identity and an **associative, stateless** function; 3-arg form with combiner when the result type differs.
- **Collectors** — `toMap` (**needs a merge function** for duplicate keys), `joining`, `groupingBy` + downstream (SQL `GROUP BY`), `partitioningBy` (always both keys), `teeing` (12), `collectingAndThen`.
- **Primitive streams** — `IntStream` / `LongStream` / `DoubleStream`: `sum`, `average` → `OptionalDouble`, `summaryStatistics`, `range`; convert with `mapToInt` / `boxed` / `mapToObj`.
- **Parallel** — Spliterator splits the source, chunks run on the **common** `ForkJoinPool` (+ caller thread). Only for big, CPU-bound, splittable, stateless work; never block; measure.
- **Gatherers** (Java 24) — `stream.gather(g)`: custom stateful intermediate ops; built-ins `windowFixed`, `windowSliding`, `scan`, `fold`, `mapConcurrent` (virtual threads, order preserved).

Key facts & numbers

On an empty stream	Result
`anyMatch` / `allMatch` / `noneMatch`	`false` / `true` / `true`
`IntStream.sum()` / `count()`	`0`
`average()`, `min()`, `max()`	empty `Optional*`
`reduce(op)` (no identity)	`Optional.empty()`

	`toList()` (16)	`Collectors.toList()`	`toUnmodifiableList()` (10)
Mutable	no	unspecified (`ArrayList` today)	no
Nulls	allowed	allowed	NPE

Stream	Loop
Clear transform/filter/aggregate steps, grouping, possible parallelism	Complex `break` / `continue`, several accumulators, checked exceptions, hottest primitive loops

Code to remember

```
Map<String, Long> freq = words.stream()
        .collect(groupingBy(Function.identity(), counting()));
Map<Dept, Double> avg = emps.stream()
        .collect(groupingBy(Emp::dept, averagingDouble(Emp::salary)));
Map<String, Emp> byId = emps.stream()
        .collect(toMap(Emp::id, e -> e, (a, b) -> a));          // merge fn!
List<String> top3 = emps.stream()
        .sorted(comparingDouble(Emp::salary).reversed())
        .limit(3).map(Emp::name).toList();
try (Stream<String> lines = Files.lines(path)) { … }            // close I/O streams
```

In the interview

Q: What is a stream and how does it execute? **A:** A lazy, single-use pipeline; nothing runs until the terminal op, elements flow one at a time, short-circuiting ops stop early so infinite streams work with `limit`.

Q: `map` vs `flatMap`? **A:** 1→1 vs 1→many, flattening nested streams.

Q: Intermediate vs terminal? **A:** Intermediate returns a new lazy stream; terminal produces a result/side effect and consumes the stream.

Q: Stateful intermediate op? **A:** One that must see other elements — `sorted`, `distinct`, `limit`, `skip` — and may buffer.

Q: Duplicate keys in `toMap`? **A:** `IllegalStateException` unless you pass a merge function.

Q: When do parallel streams help? **A:** Large, CPU-bound, independent work on splittable sources (arrays, ranges); they hurt for small inputs, `LinkedList`/`iterate` sources, ordered ops, and blocking (which starves the common pool).

Q: Why can `parallel().forEach(list::add)` fail? **A:** `ArrayList` isn't thread-safe; use `collect`/`toList`.

Q: Streams faster than loops? **A:** Not inherently; choose for clarity; parallel can win on big CPU-bound work.

Gotchas

- `Stream.of(intArray)` → a stream of one `int[]`; use `Arrays.stream`/`IntStream.of`.
- Side effects in `forEach`/`map` to build results (EJ 46) → `collect`/`toList`.
- `sorted()`/`distinct()` before `limit` on an infinite stream → hangs; `Stream.generate(() -> 1).distinct().limit(2)` hangs.
- Logic in `peek` → `count()` may skip the pipeline (Java 9+).
- `sorted().findFirst()` for a minimum → `min(comparator)`.
- Mutating a `toList()` result → `UnsupportedOperationException`; use `toCollection(ArrayList::new)`.
- `toMap` with `null` values → NPE.
- `groupingBy` order → unspecified unless you pass a map factory (`TreeMap::new`).
- Returning a `Stream` that callers iterate twice → return a `Collection` (EJ 47).
- Not closing `Files.lines`/`Files.walk` → file-handle leak.

TL;DR

Source → lazy intermediates → one terminal op, single use. `groupingBy` + downstream and `toMap` with a merge function cover most interview tasks; keep lambdas side-effect free; parallel only after measuring.

13. I/O and NIO

`java.io` vs NIO.2, `Path` and `Files`, byte vs character streams, buffering, charsets, channels, serialization, properties.

Must-know

- **Three APIs** — `java.io` streams (1.0, blocking), NIO channels/buffers/selectors (1.4), **NIO.2** `Path` / `Files` (7) — use NIO.2 for files.
- `Path` — just a name: `resolve` appends (absolute argument wins), `relativize` gives the route, `normalize` is purely textual; `toRealPath()` touches the disk.
- `Files` — specific exceptions (`NoSuchFileException`, `FileAlreadyExistsException`, `DirectoryNotEmptyException`); `copy` / `move` need `REPLACE_EXISTING`; `walk` is depth-first and includes the start dir.
- **Byte streams** (`InputStream` / `OutputStream`) for raw data vs **character streams** (`Reader` / `Writer`) for text, bridged by `InputStreamReader` / `OutputStreamWriter` **with a charset**. Wrapping = **Decorator** pattern.
- `read()` **returns** `int` — 0–255 or `-1` at end of stream.
- **Buffering** — ~100× fewer system calls; `close()` flushes; `flush()` hands data to the OS, **not** to disk (`FileChannel.force(true)` for durability).
- **Charsets** — UTF-8 default everywhere since **Java 18 (JEP 400)**; still pass `StandardCharsets.UTF_8`. Wrong charset = mojibake.
- **Serialization** — `Serializable`, `transient` fields skipped, `serialVersionUID` for compatibility, constructors not called (records excepted). **Avoid**: deserializing untrusted data can lead to remote code execution; use JSON/protobuf, or records + `ObjectInputFilter` allow-list.
- `ByteBuffer` cycle: fill → `flip` → drain → `clear`. Memory-mapped files (`FileChannel.map`) for huge random access; `transferTo` for zero-copy.

Key facts & numbers

Task	Method	Note
Small text file	`Files.readString` / `writeString` (11)	whole file in memory
Huge text file	`Files.lines` (lazy, **close it**) / `newBufferedReader`	constant memory
Append	`StandardOpenOption.APPEND`	writes **truncate** by default
Atomic create	`CREATE_NEW`	fails if exists
Crash-safe replace	temp file in same dir → `move(tmp, target, ATOMIC_MOVE, REPLACE_EXISTING)`	
Compare contents	`Files.mismatch` (12) → −1 if identical	
Delete a tree	`walk` + reverse sort, or `walkFileTree` `postVisitDirectory`	children first

Pitfall	Correct idea
`Files.exists` false	means "doesn't exist" **or** "can't tell"; `notExists` isn't its negation
`Properties.load(InputStream)`	ISO-8859-1 → use `load(Reader)` with UTF-8
`PrintWriter` / `PrintStream`	swallow `IOException` (`checkError()`)
`é` in UTF-8	2 bytes
Static fields	not serialized

Code to remember

```
Path target = uploads.resolve(name).normalize();
if (!target.startsWith(uploads)) throw new SecurityException("path traversal");

long errors;
try (Stream<String> lines = Files.lines(log, StandardCharsets.UTF_8)) {
    errors = lines.filter(l -> l.contains("ERROR")).count();   // 50 GB ok
}
```

In the interview

Q: Byte vs character streams? **A:** Bytes for raw data (PNG, zip), characters for text; bridge with an explicit charset; features stack by wrapping (Decorator); close the outermost wrapper.

Q: Count lines in a 50 GB log? **A:** `Files.lines` in try-with-resources or a `BufferedReader` loop — never `readAllLines`.

Q: Crash-safe config update? **A:** Write a temp file in the same directory, `force` if durability matters, then `ATOMIC_MOVE` over the old file.

Q: NIO vs IO? **A:** `java.io` = blocking streams; NIO = channels/buffers/selectors (non-blocking sockets, mmap); NIO.2 = `Path` / `Files`.

Q: Why avoid Java serialization? **A:** Untrusted deserialization → RCE; brittle across versions; bypasses constructors. Use JSON/protobuf.

Q: `Files.delete` vs `File.delete`? **A:** `Files.delete` throws a descriptive exception; `File.delete` just returns `false`.

Gotchas

- `dir + "/" + name` → use `resolve`.
- Not closing `Files.list/walk/find/lines` → "Too many open files".
- Check-then-act (`if (!exists) createFile`) → race; just `createFile` and catch.
- `FileReader` without a charset on ≤ 17 → platform default.
- Ignoring `read(buf)`'s count → garbage at the end of the copy.
- Forgetting `flush()` on a socket writer → peer waits forever.
- Closing a `Scanner(System.in)` → closes `System.in` for good.
- UTF-8 BOM () at file start → first CSV header mismatch.
- New `ObjectMapper` per request → expensive; share one.

TL;DR

`Path` + `Files` for files, streams/readers for data from anywhere, always an explicit charset, always try-with-resources, buffer small I/O, and don't use Java serialization for new code.

14. Concurrency I: Threads, Locks and the Toolkit

Threads and their lifecycle, interruption, races, `synchronized`, `volatile`, `wait` / `notify`, locks, atomics, executors, concurrent collections, synchronizers.

Must-know

- **Threads** share the heap, own their stack. **`start()`** creates a new thread; calling `run()` just runs it on the current thread. `join()` waits. **Daemon** threads don't keep the JVM alive. Prefer `Runnable`/lambda over `extends Thread`.
- **Concurrency vs parallelism** — tasks in progress over the same period vs literally at the same instant on different cores.
- **Race conditions** — **read-modify-write** (`count++`) and **check-then-act** (`if (!map.containsKey) put`); results vary by run; tests can't prove absence.
- **`synchronized`** — mutual exclusion **and** visibility; locks `this` (instance method), the `Class` object (static method) or a chosen object; **reentrant**; released even on exceptions. Every access to the state must use the **same lock**.
- **`volatile`** — visibility and ordering only; perfect for stop flags and publishing immutable snapshots; `volatile++` still races.
- **Interruption** — cooperative: `interrupt()` sets a flag; blocking calls throw `InterruptedException` **and clear the flag** → rethrow or `Thread.currentThread().interrupt()`. (`Thread.stop` was removed: it released locks mid-operation.)
- **`wait` / `notify`** — hold the lock, wait in a **`while`** loop (spurious wake-ups), prefer `notifyAll`; prefer higher-level tools.

- **`ReentrantLock`** — `tryLock` (with timeout), `lockInterruptibly`, fairness, multiple `Condition`s; **unlock in `finally`**. `ReadWriteLock` for read-heavy data; `StampedLock` for optimistic reads.
- **Atomics** — CAS-based, lock-free, single variable; `AtomicReference` + immutable object; **`LongAdder`** for hot counters (`sum()` isn't an atomic snapshot).
- **Executors** — tasks as `Runnable`/`Callable`, results/exceptions via `Future` (`ExecutionException`); always shut down (or try-with-resources `close()`, Java 19).
- **`ConcurrentHashMap`** — atomic `putIfAbsent`/`computeIfAbsent`/`compute`/`merge`; lock-free reads; CAS into empty bins + lock on a bin's first node (Java 8; Java 7 used segments); **no null keys/values**; weakly consistent iterators.
- **Thread-safety strategies** (JCIP) — confinement, immutability, encapsulated synchronisation, delegation to thread-safe components.

Key facts & numbers

Thread state	Cause
`NEW`	created, not started
`RUNNABLE`	running or ready (even blocked in OS I/O)
`BLOCKED`	waiting for a **monitor** (`synchronized`)
`WAITING`	`wait()`, `join()`, `LockSupport.park()` (locks, latches, queues)
`TIMED_WAITING`	`sleep`, `wait(ms)`, `join(ms)`, `tryLock(timeout)`
`TERMINATED`	`run()` finished

	`volatile`	**`synchronized`**	**`AtomicInteger`**
Visibility	✓	✓	✓
Atomic `++`	✗	✓	✓
Multi-variable invariants	✗	✓	✗
Blocks	never	may	never (retries)

Factory	Queue	Risk
`newFixedThreadPool(n)`	**unbounded** `LinkedBlockingQueue`	tasks pile up → OOM
`newCachedThreadPool()`	`SynchronousQueue`; idle threads die after 60 s	unbounded threads
`newSingleThreadExecutor()`	unbounded	sequential
`newScheduledThreadPool(n)`	delay queue	periodic task dies silently on exception
`newWorkStealingPool()` (8)	ForkJoinPool	many small CPU tasks
`newVirtualThreadPerTaskExecutor()` (21)	none	not a pool

Synchronizer	Idea	Reusable
`CountDownLatch(n)`	await until n `countDown()`s	no
`CyclicBarrier(n)`	n threads wait for each other	yes
`Semaphore(k)`	k permits — limit concurrency	yes
`Phaser` (7)	dynamic parties, numbered phases	yes

Concurrent collection	Use
`ConcurrentSkipListMap/Set`	sorted concurrent, O(log n)
`CopyOnWriteArrayList`	read-mostly (listeners); snapshot iterators, O(n) writes
`ArrayBlockingQueue`	bounded producer-consumer (`put` / `take` block)
`LinkedBlockingQueue`	unbounded by default
`SynchronousQueue`	capacity 0 hand-off
`ConcurrentLinkedQueue`	lock-free, `poll` → `null`

- **Pool sizing** (JCIP): CPU-bound ≈ N_{cpu} (+1); waiting tasks ≈ N_{cpu} × (1 + wait/compute) — 8 cores, 90 ms wait per 10 ms CPU → ~80 threads. Then measure. Virtual threads usually remove this for I/O-bound work.

Code to remember

```
private final Object lock = new Object();
private int count;
void inc() { synchronized (lock) { count++; } }
int get()  { synchronized (lock) { return count; } }          // reader too!

lock.lock();
try { /* critical section */ } finally { lock.unlock(); }

try (var pool = Executors.newFixedThreadPool(4)) {             // Java 19+ close()
    Future<Integer> f = pool.submit(() -> compute());
    int r = f.get(1, TimeUnit.SECONDS);                        // ExecutionException wraps failures
}
counts.merge(word, 1, Integer::sum);                           // atomic on ConcurrentHashMap
```

In the interview

Q: `synchronized` vs `volatile` vs `AtomicInteger`? **A:** Mutual exclusion + visibility / visibility only / lock-free atomic single-variable updates.

Q: Do static and instance `synchronized` methods block each other? **A:** No — different locks (`Class` object vs instance).

Q: How do you handle `InterruptedException`? **A:** Rethrow it, or restore the flag with `Thread.currentThread().interrupt()`; never swallow it.

Q: When `ReentrantLock` over `synchronized`? **A:** When you need `tryLock`/timeouts (deadlock avoidance), interruptible acquisition, fairness or several `Condition`s; otherwise `synchronized`.

Q: `shutdown` vs `shutdownNow`? **A:** Graceful (queued tasks run) vs interrupt running tasks and return queued ones.

Q: `submit` vs `execute`? **A:** `submit` returns a `Future` and captures exceptions; `execute` sends them to the uncaught handler.

Q: Production thread pool config? **A:** Explicit `ThreadPoolExecutor`: bounded queue, sized max, rejection policy such as `CallerRunsPolicy` (back-pressure), named threads.

Q: Why does `ConcurrentHashMap` reject nulls? **A:** `get` returning `null` must unambiguously mean absent; a follow-up `containsKey` isn't safe under concurrency.

Q: Make seat booking safe? **A:** Atomic check-and-book: `synchronized`/per-seat lock or `putIfAbsent(seatId, userId)`; across servers, a DB constraint or `SELECT … FOR UPDATE`.

Gotchas

- Calling `run()` instead of `start()` → no new thread.
- Synchronising the writer but not the reader → stale reads.
- Non-volatile `boolean` stop flag → the loop may never see the change.

- `Collections.synchronizedList` iteration → still needs `synchronized (list) { … }`.
- `containsKey` + `put` on `ConcurrentHashMap` → check-then-act race; use atomic methods.
- Slow/blocking work inside `computeIfAbsent` → locks that bin; never modify the same map inside.
- `ConcurrentHashMap.size()` under load → only an estimate.
- Forgetting to shut down a pool → JVM never exits.
- Tasks waiting on tasks in the same bounded pool → thread-starvation deadlock.
- Ignoring `Future`s → exceptions vanish.
- `setPriority` for correctness → only a hint.
- Interrupting classic socket reads → doesn't unblock; close the socket.

TL;DR

All shared mutable state needs one consistent lock, an atomic, or a thread-safe container. `volatile` is visibility only; restore interrupts; prefer executors with bounded queues; `ConcurrentHashMap` + `merge`/`computeIfAbsent`.

15. Concurrency II: Modern Concurrency

`Future` and `CompletableFuture`, Fork/Join, virtual threads and pinning, `ThreadLocal` vs scoped values, structured concurrency, the Java Memory Model, liveness.

Must-know

- **`Future`** — blocking `get()`, no composition (`state()`/`resultNow()` since 19).
- **`CompletableFuture`** — composable, completable future: `supplyAsync(…, executor)` → `thenApply` (map) / `thenCompose` (flatMap) / `thenCombine` (zip) / `allOf` / `anyOf`; errors via `exceptionally` / `handle` / `whenComplete`; latency via `orTimeout` / `completeOnTimeout` (9). Async variants default to the **common pool** → pass your own executor for I/O.
- **Fork/Join** — recursive split + **work stealing** (`RecursiveTask`): fork one half, compute the other, join; powers parallel streams; CPU-bound only.
- **Virtual threads** (final **21**) — cheap JVM-managed threads, **M:N** on carrier threads (≈ one per core); a blocked virtual thread **unmounts** and frees its carrier → millions of waiting tasks. More throughput for waiting work, not faster per task. **Don't pool them**; limit scarce resources with a `Semaphore`/bounded connection pool. Always daemon.
- **Pinning** — Java 21–23: blocking inside `synchronized` pinned the carrier; **Java 24 (JEP 491) fixed it**; native callbacks and class initialisation still pin; JFR event `jdk.VirtualThreadPinned`.
- **`ThreadLocal`** — per-thread mutable copy; leaks across pooled tasks unless `remove()` in `finally`; a million copies with virtual threads. **`ScopedValue`** (final **25**) — immutable binding for the duration of a call, rebind only for nested calls, inherited by structured subtasks; use for request context.
- **Structured concurrency** — `StructuredTaskScope`: subtasks' lifetime bound to a block, sibling cancellation on failure, no leaked threads. **Still preview**: 25 (JEP 505), 26 (JEP 525), 27 (JEP 533).
- **Java Memory Model / happens-before** — program order; unlock → later lock of the **same** monitor; volatile write → later read of **that** field; `start()`; `join()`; interrupt; `java.util.concurrent` hand-offs; transitive. No edge = **data race** (stale/reordered reads).
- **Safe publication / `final`** — final fields are correctly visible after construction if `this` doesn't escape; immutable objects are always safe to share.
- **Liveness** — **deadlock** (Coffman: mutual exclusion, hold-and-wait, no preemption, circular wait), **livelock** (busy retrying in lock-step), **starvation** (never gets the resource).

Key facts & numbers

Error handler	Runs	Changes result?	Use
`exceptionally`	on failure	yes	fallback value
`handle`	always	yes	map success + failure
`whenComplete`	always	no	logging, metrics
`exceptionallyCompose` (12)	on failure	yes, async	retry / backup service

	Platform thread	Virtual thread
Mapping	1:1 OS thread	M:N on carriers
Cost	1–2 MB stack reserved, OS call	small heap object
How many	thousands	millions
Blocking	holds the OS thread	unmounts
Pool?	yes	**no**

Failure	Threads	CPU	Fix
Deadlock	`BLOCKED` / `WAITING`	~0%	global lock order, `tryLock` + timeout, no foreign calls under locks
Livelock	running	high	randomised back-off
Starvation	runnable, unlucky	—	fairness, separate pools (bulkheads)

Workload	Tool (Java 25)
Many blocking calls (HTTP, DB)	virtual threads, thread-per-request
Fan-out and combine	`CompletableFuture.allOf` (structured concurrency once final)
CPU-heavy in-memory work	parallel streams / Fork/Join / ≈ #cores platform pool
Periodic jobs	`ScheduledExecutorService`
Shared counters/caches	atomics, `LongAdder`, `ConcurrentHashMap`

Code to remember

```
try (var exec = Executors.newVirtualThreadPerTaskExecutor()) {
    var user  = CompletableFuture.supplyAsync(() -> api.user(id), exec);
    var order = CompletableFuture.supplyAsync(() -> api.orders(id), exec);
    Page p = user.thenCombine(order, Page::new)
                 .orTimeout(2, TimeUnit.SECONDS)
                 .exceptionally(ex -> Page.empty())
                 .join();
}
class Holder { static final Heavy INSTANCE = new Heavy(); }   // lazy, thread-safe
private volatile Heavy h;                                     // DCL needs volatile
Heavy get() { var r = h; if (r == null) synchronized (this) { r = h; if (r == null) h = r = new Heavy(); } return
r; }
```

In the interview

Q: `thenApply` vs `thenCompose`? **A:** `map` vs `flatMap` — use `thenCompose` when the function returns a `CompletableFuture`.

Q: `join` vs `get`? **A:** Same result; `join` throws unchecked `CompletionException`, `get` throws checked `ExecutionException` / `InterruptedException` and has a timeout overload.

Q: Wait for a list of futures? **A:** `allOf(...).thenApply(v -> list.stream().map(CompletableFuture::join).toList())`.

Q: Explain virtual threads. **A:** JVM threads multiplexed onto a few carriers; blocking unmounts them, so thread-per-request code scales like async code; for I/O-bound work, not CPU-bound; one per task, no pooling.

Q: What replaces a pool-size limit with virtual threads? **A:** A `Semaphore` or bounded resource pool (e.g. HikariCP).

Q: `ThreadLocal` vs scoped values? **A:** ThreadLocal is mutable, per-thread, leak-prone with pools and costly with virtual threads; ScopedValue is immutable, bounded to a call, shared with structured subtasks.

Q: What is happens-before and why does DCL need `volatile`? **A:** The JMM rule that guarantees visibility; without `volatile` another thread can see a non-null reference to a not-yet-constructed object.

Q: Does `volatile` make `i++` atomic? **A:** No — visibility and ordering only.

Q: Prevent and find deadlocks? **A:** Global lock ordering or `tryLock` with timeouts; `jstack <pid>` / `jcmd <pid> Thread.print` reports Java-level deadlocks.

Gotchas

- Blocking I/O in `supplyAsync` without an executor → exhausts the common pool, stalls every parallel stream.
- `join()` right after every `supplyAsync` → synchronous code with extra steps.
- `thenApply` returning a future → `CompletableFuture<CompletableFuture<T>>`.
- `cancel(true)` on a `CompletableFuture` → completes with `CancellationException`; the work keeps running.
- `exceptionally(ex -> null)` → swallowed errors.
- Pooling virtual threads / heavy `ThreadLocal` caches with them.
- Assuming downstream systems scale with virtual threads → the DB still handles ~200 at a time.
- `volatile` reference to a mutable object → only the reference write is covered.
- "It worked in testing" → reordering depends on CPU (ARM reorders more than x86), JIT and load.

TL;DR

I/O-bound → virtual threads (no pooling, semaphores for limits); CPU-bound → Fork/Join/parallel streams; fan-out → `CompletableFuture` with your own executor. Visibility needs a happens-before edge; prevent deadlock with lock ordering or `tryLock`.

Build-it-yourself problems (Chapter 22)

- **Bounded buffer**: ring buffer + `while` guards; monitor + `notifyAll`, two `Condition`s + `signal`, or `freeSlots`/`filledSlots` semaphores before the mutex; busy-wait is correct but burns a core.
- **Semaphore / RW lock / barrier**: `while (permits == 0) wait()`; writer preference via `waitingWriters` (undo on interrupt); barrier waits on a **generation**, not the count.
- **Deferred callbacks**: min-heap by `nanoTime` deadline + `awaitNanos(head − now)`; signal only for a new head.
- **Async → sync**: per-call state + timed wait, or `CompletableFuture.get(timeout)`; never block the callback thread.
- **Ordered printing / token bucket**: gates that remember (`CountDownLatch`, `Semaphore(0)`); lazy refill from elapsed time, a refiller thread started by a factory, not the constructor.
- **Puzzles** (unisex bathroom, Uber ride, sleeping barber): one predicate under one lock, `notifyAll` when waiters differ, batching against starvation.

16. JVM Internals and Performance

Class loading, memory areas, object layout, GC algorithms and collectors, tuning, JIT and escape analysis, JMH, diagnostics, leaks, modules.

Must-know

- **Class loading** — lazy: **load** (find bytes) → **link** (verify, prepare statics with defaults, resolve lazily) → **initialise** (static initialisers once, textual order, under a lock) on first active use.
- **Parent delegation** — bootstrap (`getClassLoader()` returns `null`) → platform → application → custom; core classes can't be spoofed, each class loaded once per hierarchy. **CDS** = memory-mapped archive of pre-processed classes (faster start-up).
- `ClassNotFoundException` (checked; `Class.forName` can't find a name) vs `NoClassDefFoundError` (present at compile time, missing or failed to initialise at run time).
- **GC basics** — mark (cost ∝ live objects), sweep, compact, copy/evacuate; **generational** because most objects die young (needs card tables/remembered sets); concurrent collectors need **barriers**.
- **G1** (default since 9, JEP 248) — equal regions (1–32 MB), young evacuated in STW pauses, old marked concurrently, **mixed** collections take the most-garbage regions first; objects ≥ half a region are **humongous**; pause goal `MaxGCPauseMillis` 200 ms.
- **ZGC** — concurrent relocation via **coloured pointers** + **load barriers**; pauses well under 1 ms regardless of heap (up to 16 TB); generational (21), generational-only (24).
- **Shenandoah** — concurrent compaction with load-reference barriers; not in Oracle builds.
- **JIT (C2)** — inlining (hot methods ≤ 325 bytecode bytes, others ≤ 35), devirtualisation, **escape analysis** → scalar replacement (no allocation) and lock elision, loop optimisations, dead-code elimination, speculation + **deoptimisation**.
- **Benchmarking** — naive loops lie (warm-up, dead-code elimination, constant folding, GC noise, profile pollution); **JMH** gives warm-up/measurement iterations, forked JVMs, `Blackhole.consume`.
- **Memory leak** — objects still reachable but never used; heap **after GC** keeps growing (high usage plateaus).
- **JPMS** — `requires` (module), `exports` (package), `opens` (deep reflection at run time); reliable configuration + strong encapsulation; class path = **unnamed module**; `jdeps` analyses, `jlink` builds a small runtime (44 MB vs 377 MB JDK in the book's example).

Key facts & numbers

Area	Holds	Flag	Error
Heap (shared)	objects, arrays	`-Xms`, `-Xmx`, `-XX:MaxRAMPercentage` (default 25)	`OOM: Java heap space` / `GC overhead limit exceeded`
Java stack (per thread)	frames	`-Xss` (~1 MB, 2 MB on macOS/AArch64)	`StackOverflowError`
Metaspace (native)	class metadata	`-XX:MaxMetaspaceSize` (unlimited)	`OOM: Metaspace`
Code cache	JIT output	`-XX:ReservedCodeCacheSize`	"CodeCache is full" — runs slowly
Direct/native	direct buffers, thread stacks	`-XX:MaxDirectMemorySize`	`OOM: Direct buffer memory` / `unable to create native thread`

Object (64-bit, compressed oops)	Default	Compact headers (25)
`new Object()`	16 B	8 B
`record Point(int x, int y)`	24 B	16 B
`new int[10]`	56 B	56 B
6-char `String`	48 B	48 B

Header 12 B (mark word 8 + compressed class pointer 4), references 4 B (heap < ~32 GB), 8-byte alignment.

Collector	Pauses	Choose for
Serial	grow with heap (~130 ms in the book's test)	tiny heaps, 1 CPU, CLI
Parallel	shorter, still grow	batch throughput
G1	few–tens of ms (~9 ms)	general default
ZGC	< 1 ms (~0.01 ms)	latency, huge heaps
Shenandoah	sub-ms to few ms	low latency
Epsilon	none — exits when full	perf tests

Symptom	First tool
App hangs	`jcmd <pid> Thread.print` 2–3× (look for `BLOCKED`, "Found one Java-level deadlock")
High CPU	JFR hot methods / async-profiler flame graph
Latency spikes	GC log pauses; JFR gc-pauses, safepoints
Memory growth	`jstat -gcutil`; `GC.class_histogram` twice; JFR `OldObjectSample`; heap dump (MAT)
RSS > heap	NMT `VM.native_memory`

- Useful flags: `-Xlog:gc*`, `-XX:+HeapDumpOnOutOfMemoryError`, `-XX:+AlwaysPreTouch`, `-XX:+UseCompactObjectHeaders` (25), `-XX:+DisableExplicitGC`.

Code to remember

```
java -XX:+UseZGC -Xmx4g -XX:MaxRAMPercentage=75 -Xlog:gc*:file=gc.log -jar app.jar
jcmd <pid> Thread.print              # thread dump (= jstack)
jcmd <pid> GC.class_histogram        # who fills the heap
jcmd <pid> JFR.start duration=60s filename=rec.jfr
jstat -gcutil <pid> 1000             # live GC counters every second
```

In the interview

Q: What happens when the JVM first uses a class? **A:** Loaded via parent delegation, linked (verified, statics prepared, references resolved lazily), then initialised once under a lock.

Q: Compare G1 and ZGC. **A:** Both region-based and generational. G1 evacuates in STW pauses aimed at a goal with lower overhead; ZGC relocates concurrently (coloured pointers, load barriers) for sub-ms pauses at some throughput and headroom cost.

Q: Why do concurrent collectors need barriers? **A:** The app changes references while the GC traces or moves objects; barriers record or repair those changes.

Q: What is escape analysis? Does Java allocate on the stack? **A:** C2 proves an object never escapes and replaces it with its fields (no allocation, no locks). Not literally stack allocation — scalar replacement has the same effect; depends on inlining.

Q: Latency spikes every few minutes? **A:** Correlate with GC log (long pauses, full GCs, evacuation failures), record JFR across a spike (GC, safepoints, locks, allocation); fix allocation rate or headroom before touching flags.

Q: Can Java leak memory? **A:** Yes — static maps, listeners never removed, `ThreadLocal`s on pooled threads, mutable map keys, classloader leaks. Confirm heap-after-GC growth → histogram → path to GC root.

Q: Thread dump vs heap dump? **A:** Thread dump: stacks and locks, small and fast. Heap dump: every object, large, pauses the JVM, may contain secrets.

Q: `exports` vs `opens`? **A:** Compile/run-time access to public types vs run-time reflective access to all members.

Gotchas

- Sizing a container by `-Xmx` alone → budget metaspace, code cache, thread stacks, direct buffers, GC overhead.
- Huge `-Xss` to "fix" `StackOverflowError` → fix the recursion.
- `-Xmn` / `NewRatio` with G1 → disables pause-time adaptation.
- `System.gc()` → triggers a full GC.
- Measuring GC on a laptop, deploying to a 2-CPU container.
- Heavy static initialisers → `ExceptionInInitializerError`, then `NoClassDefFoundError` forever.
- `WeakReference` for caches → use size/time eviction (Caffeine).
- Forgetting `opens` for reflective frameworks → `InaccessibleObjectException`; split packages on the module path → error.
- `System.currentTimeMillis()` for intervals → `System.nanoTime()`.

Must-know — OS fundamentals (§16.13)

- **Process vs thread** — process = own virtual address space + fd table; threads share the process's memory. Platform thread = 1 OS thread; virtual threads = M:N on carrier threads. `fork()` uses copy-on-write; `ProcessBuilder` = fork + exec.
- **Scheduling** — preemptive time slices, context switches cost µs + cache warm-up; cgroup CPU quotas throttle the whole JVM. 100% CPU → `top -H` thread id → hex → `nid=` in `jcmd <pid> Thread.print`.
- **Virtual memory** — pages (4 KiB x86 Linux, 16 KiB Apple Silicon), page table + TLB; allocation is lazy (page fault on first touch) → VSZ ≫ RSS; major fault = disk (ms). Out of RAM → swap or **OOM killer** (`SIGKILL`, no `OutOfMemoryError`).
- **Syscall** — user → kernel mode trap (~100 ns–µs); batch I/O with buffers.
- **File descriptor** — per-process integer handle for files/sockets/pipes; limit `ulimit -n`; leak → `Too many open files`.
- **I/O models** — blocking, non-blocking, multiplexing (`select` / `poll` O(n), **epoll/kqueue** O(ready) → NIO `Selector`, Netty, Redis), completion (io_uring), or virtual threads over blocking code.
- **Page cache** — file data cached in spare RAM; `write` only dirties pages, durability needs `fsync` (`FileChannel.force`); `transferTo` = zero-copy `sendfile`.

Level	Latency
L1 / DRAM	~1 ns / ~100 ns
NVMe read / disk seek	~20–100 µs / ~5–10 ms
Same-DC / cross-continent RTT	~0.5 ms / ~50–150 ms

Q: Process vs thread? **A:** A process has its own address space and resources; threads share their process's memory, so they are cheaper but need synchronisation.

Q: epoll vs select? **A:** `select` rescans every fd per call; `epoll` registers once and returns only ready fds, so one thread can watch tens of thousands of sockets.

Q: Container JVM disappeared with no stack trace? **A:** The OOM killer: total RSS (heap + metaspace + stacks + direct memory) exceeded the cgroup limit; lower `-Xmx` / `MaxRAMPercentage`.

TL;DR

Heap/stack/metaspace/code cache each fail differently; G1 is the default, ZGC for sub-ms pauses; C2 inlines and scalar-replaces non-escaping objects; measure with JMH and JFR, diagnose with `jcmd`.

17. Modern Java: 17 to 25

Reading a feature's status, what each release finalised from 17 to 25, what is still preview, and upgrade breakages.

Must-know

- **Status words** — **final** = use it; **preview** (`--enable-preview` on `javac` and `java`), **incubator** (`jdk.incubator.*`), **experimental** (`-XX:+UnlockExperimentalVMOptions`) = experiments only; **withdrawn** = gone. Check the JEP title on openjdk.org.
- **Java 17 baseline** — switch expressions (14), helpful NPEs (14), text blocks (15), records (16), `instanceof` patterns (16), `Stream.toList()` (16), sealed classes (17).
- **21 LTS** — the big one: **virtual threads (444)**, pattern matching for `switch` (441), record patterns (440), sequenced collections (431), generational ZGC (439).
- **25 LTS** — compact source files + instance `main` + `IO` (512), `import module` (511), flexible constructor bodies (513), **scoped values** (506), KDF API (510), compact object headers opt-in (519), generational Shenandoah (521), AOT ergonomics (514/515).
- **Still preview/incubating in 25** — structured concurrency (5th preview, 505), primitive types in patterns (3rd preview, 507), stable values (preview 502 → renamed lazy constants), PEM encodings (470), Vector API (10th incubator, 508).
- **String templates** — previewed in 21 and 22, **withdrawn in 23**; no replacement yet.

Key facts & numbers

Release	Headline finals
17 LTS (2021-09)	sealed classes, `RandomGenerator` (356), strong encapsulation of JDK internals (403)
18	**UTF-8 by default** (400), `jwebserver`, Javadoc `@snippet`; finalization deprecated (421)
19, 20	previews/incubators only (RISC-V port in 19)
21 LTS (2023-09)	virtual threads, pattern `switch`, record patterns, sequenced collections, generational ZGC
22	unnamed `_` variables/patterns (456), **FFM API** (454), multi-file source launch (458)
23	Markdown doc comments (467); ZGC generational by default; string templates withdrawn
24	**stream gatherers** (485), Class-File API (484), **no** `synchronized` **pinning** (491), AOT class loading (483), Security Manager disabled (486), ML-KEM/ML-DSA
25 LTS (2025-09-16)	compact source/instance `main`, `import module`, flexible constructors, scoped values, KDF, compact headers

Upgrade breakage	Fix
Strong encapsulation (17)	upgrade the library; `--add-opens` temporarily
Security Manager disabled (24)	remove `System.setSecurityManager`
`finalize()` deprecated (18)	try-with-resources / `Cleaner`
Dynamic agent loading warns (21)	`-javaagent` at start-up (e.g. Mockito)
`Unsafe` memory access warns (24)	upgrade Netty/serializers
JNI warns (24)	`--enable-native-access=…`
`-XX:-ZGenerational` removed (24)	drop the flag
String-template code	rewrite

Code to remember

```java
// Java 25 compact source file — no class, no package
void main() {
    String name = IO.readln("Name? ");
    IO.println("Hello, " + name);
}
// Java 25 flexible constructor body
Employee(String name, int age) {
    if (age < 18) throw new IllegalArgumentException("age");   // before super(), no `this`
    super(name);
}
```

In the interview

Q: What changed 17 → 21 and 21 → 25? **A:** 17→21: pattern `switch`, record patterns, virtual threads, sequenced collections, generational ZGC. 21→25: `_`, FFM API, gatherers, Class-File API, no `synchronized` pinning, then compact source files, module imports, flexible constructors, scoped values, AOT start-up.

Q: Is structured concurrency final? **A:** No — 5th preview in 25, 7th in 27.

Q: Why upgrade 21 → 25? **A:** Longer LTS horizon, no pinning with virtual threads, faster start-up (AOT caches), smaller heaps (compact headers), new conveniences.

Q: Preview features in production? **A:** No — they may change or disappear, and class files compiled with previews run only on that exact release.

Q: Which release finalised virtual threads? **A:** Java 21, JEP 444.

Q: What does `import module java.base;` import? **A:** All public top-level types of every package `java.base` exports.

Gotchas

- Using a 21 preview feature (instance `main`, string templates) as if final.
- Assuming 19/20 finalised language features → previews only.
- Code touching JDK internals by reflection → breaks since 17.
- Upgrading without running tests on the new JDK first.

TL;DR

17 = sealed + records baseline; 21 = virtual threads + pattern matching; 24 = gatherers + no pinning; 25 = compact sources, module imports, flexible constructors, scoped values. Structured concurrency and primitive patterns are still preview.

18. Testing, Build and Tooling

Project layout, Maven and Gradle, JUnit, Mockito, AssertJ, test design, logging, JDBC, packaging.

Must-know

- **Layout & coordinates** — `src/main/java`, `src/test/java`; libraries are `groupId:artifactId:version` (SNAPSHOT = work in progress) from **Maven Central**, cached in `~/.m2/repository` / `~/.gradle/caches`; transitive dependencies come automatically.
- **Maven** — declarative `pom.xml`; fixed **lifecycle** (validate → compile → test → package → verify → install → deploy); running a phase runs all earlier ones; plugin goals bind to phases; conflicts resolved **nearest-wins** → pin in `dependencyManagement` / import a **BOM**. Use `./mvnw`.
- **Gradle** — scripted, incremental **task graph** with build cache and daemon; `implementation` (not exposed), `api` (exposed), `compileOnly`, `runtimeOnly`, `testImplementation`; **`useJUnitPlatform()`**; use `./gradlew`.
- **JUnit 5/6 (Jupiter)** — Platform (launcher) + Jupiter (API); **new test instance per method**; `@BeforeEach/@AfterEach`, static `@BeforeAll/@AfterAll`; `assertEquals(expected, actual)`, `assertThrows`, `assertAll`; `@Nested`, `@Tag`, `@TempDir`, `@ExtendWith`. JUnit 6 = Java 17 baseline, unified versions, same API.

- **Parameterized tests** — `@ParameterizedTest` + `@ValueSource` / `@CsvSource` / `@MethodSource` / `@EnumSource` / `@NullAndEmptySource`; each input reported separately; put boundaries in the table.
- **Test doubles** — dummy, **stub** (canned answers), **fake** (working simplified impl), **spy** (real object + recording), **mock** (verified interactions). Inject collaborators via the constructor; mock only true boundaries.
- **Mockito** — `@Mock`, `when(...).thenReturn(...)`, `verify`, matchers (all or none), `ArgumentCaptor`, strict stubs (`UnnecessaryStubbingException`); `doReturn` for spies, `doThrow` for void methods; `mockStatic` in try-with-resources; attach with `-javaagent` on JDK 21+.
- **AssertJ** — `assertThat(x).…` fluent, type-aware; `containsExactly`, `extracting`, `isEqualByComparingTo` (BigDecimal), `assertThatThrownBy`, soft assertions, `usingRecursiveComparison`.
- **Test design** — **FIRST** (Fast, Independent, Repeatable, Self-validating, Timely), arrange–act–assert, boundary values, test pyramid, Testcontainers for integration; coverage ≠ quality (mutation testing is a better signal).
- **Logging** — SLF4J facade + Logback/Log4j 2 backend; logger per class; `{}` placeholders (no string built when disabled); exception as last argument; MDC for request ids; no secrets; patch libraries (**Log4Shell**, CVE-2021-44228).
- **JDBC** — `DataSource` (pooled, e.g. HikariCP) → `Connection` → `PreparedStatement` (bind params, index from **1**) → `ResultSet`, all in try-with-resources; `setAutoCommit(false)` + `commit` / `rollback`; `SQLException` is checked (`getSQLState`, `23505` = unique violation).

Key facts & numbers

Maven scope	Compile	Test	Runtime/packaged	Example
`compile` (default)	✓	✓	✓	Jackson
`provided`	✓	✓	✗	Servlet API
`runtime`	✗	✓	✓	JDBC driver
`test`	✗	✓	✗	JUnit, Mockito
`import`	BOM in `dependencyManagement`			

JUnit 4	Jupiter
`@Before` / `@After`	`@BeforeEach` / `@AfterEach`
`@BeforeClass`	`@BeforeAll`
`@Ignore`	`@Disabled`
`@Test(expected=…)`	`assertThrows`
`@RunWith`, `@Rule`	`@ExtendWith`
message **first**	message **last**

Maven	Gradle
XML, fixed lifecycle, uniform	Kotlin/Groovy script, task graph, incremental, faster on big builds; Android

Level: JUL	SLF4J
SEVERE / WARNING / INFO / FINE / FINER-FINEST	ERROR / WARN / INFO / DEBUG / TRACE

Flaky cause	Cure
Current time	inject `Clock` (`Clock.fixed`)
Randomness	seeded `Random`
Order / static state	fresh state per test
`Thread.sleep`	wait for a condition (Awaitility, latches)
Network	doubles; Testcontainers/WireMock
Hash order	`containsExactlyInAnyOrder`

- Packaging: JAR = ZIP + manifest; `java -jar` needs `Main-Class`; fat JAR (Shade/Shadow), Spring Boot JAR (nested JARs), `jlink` runtime, `jpackage` installers, containers, GraalVM native image.
- Commands: `mvn clean verify` (CI), `mvn -Dtest=CartTest#method test`, `mvn dependency:tree`, `./gradlew build`.

Code to remember

```
@ExtendWith(MockitoExtension.class)
class OrderServiceTest {
    @Mock PaymentGateway gateway;                    // constructor-injected into OrderService
    @Test void chargesCustomer() {
        when(gateway.charge(anyString(), any())).thenReturn(new PaymentResult(true, "PAY-9"));
        // ... act: service.checkout("ann", cart, "SAVE25") ...
        verify(gateway).charge(eq("ann"), eq(new BigDecimal("102.50")));  // all matchers or none
    }
}
@ParameterizedTest @CsvSource({"SAVE25, 99.99, 0", "SAVE25, 100.00, 25"})   // boundaries
void percentForCode(String code, BigDecimal subtotal, int expected) {
    assertEquals(expected, Discounts.percentFor(code, subtotal));
}
```

In the interview

Q: How does Maven work? **A:** `pom.xml` gives coordinates, scoped dependencies and plugins; transitive resolution from Central into `~/.m2` (nearest wins); a fixed lifecycle where a phase runs all earlier ones.

Q: `dependencies` vs `dependencyManagement`? BOM? **A:** The first adds dependencies; the second only fixes versions/scopes. A BOM is a pom with only `dependencyManagement`, imported to align a library family.

Q: Why is `@BeforeAll` static? **A:** There is one instance per test method, so a once-per-class hook can't belong to an instance (unless `PER_CLASS` lifecycle).

Q: Mock vs stub vs spy? **A:** Stub answers; mock is also verified; spy wraps a real object and calls real methods unless stubbed.

Q: Test code using `LocalDateTime.now()`? **A:** Inject a `Clock`; use `Clock.fixed` in tests.

Q: Is 100% coverage worth it? **A:** No — diminishing returns and says nothing about assertion quality.

Q: `Statement` vs `PreparedStatement`? **A:** Prepared binds parameters separately: prevents SQL injection, typed, plan reuse.

Q: Why a connection pool? **A:** Opening a connection costs a handshake, authentication and a server process; a pool reuses a bounded number.

Q: Why a logging facade and `{}`? **A:** Libraries shouldn't force a backend; `{}` avoids building strings when the level is off.

Gotchas

- Swapped expected/actual → backwards failure messages.
- JUnit 4 `org.junit.Test` with Jupiter → silently not run; Gradle without `useJUnitPlatform()` → "0 tests", green build.
- Test libraries without `<scope>test</scope>` → shipped in the product.
- Omitting plugin versions → build changes with Maven upgrades.

- `when(spy.method())` → calls the real method; use `doReturn`.
- Mixing raw values and matchers in one call.
- `assertThat(x);` alone asserts nothing; `.as()` must come before the assertion.
- `getInt` on SQL NULL → `0`; check `wasNull()`.
- Unclosed connections → pool exhausted, app hangs.
- `double` for money columns → `BigDecimal`.
- Log-and-rethrow at every layer; logging secrets.

TL;DR

Maven/Gradle with pinned versions and wrappers; JUnit 5/6 + Mockito + AssertJ with injected collaborators and FIRST tests; SLF4J with `{}`; JDBC with pooled `DataSource`, `PreparedStatement`, try-with-resources and explicit transactions.

19. Java for Coding Interviews

The Java DSA toolkit and its traps, a complexity table, the twenty patterns at a glance, and how to talk through a solution.

Must-know

- **Test with a helper, not `assert`** — the keyword is disabled unless you run `java -ea`.
- **Toolkit** — `int[]` + `Arrays` (`sort`, `fill`, `copyOfRange`, `toString`); `c - 'a'` for letter indexes (`int[26]` frequency tables beat `HashMap<Character,Integer>`); `StringBuilder` (`append`, `reverse`, `setLength`, `deleteCharAt`); `HashMap.merge` / `getOrDefault` / `computeIfAbsent`; `if (!seen.add(x))` for duplicates.
- **`ArrayDeque`** for stacks (`push`/`pop`/`peek`) **and** queues (`offer`/`poll`/`peek`) — never `Stack` (synchronized `Vector`, exposes `get(i)`, iterates bottom-to-top). Rejects `null`.
- **`PriorityQueue`** — min-heap by default; max-heap with `Comparator.reverseOrder()`; `int[]` entries with `Comparator.comparingInt(x -> x[0])`; iteration/printing is heap order, not sorted; `contains`/`remove(obj)` O(n).
- **`TreeMap`/`TreeSet`** — `floor` (≤), `ceiling` (≥), `lower` (<), `higher` (>); return `null` when nothing qualifies; O(log n).
- **Sorting** — primitives: `Arrays.sort` (Dual-Pivot Quicksort, ascending only, no comparator); objects: stable TimSort. For `int[][]`: `Comparator.<int[]>comparingInt(x -> x[0]).thenComparingInt(x -> x[1])` (type witness needed when chaining).
- **Map keys** — need value-based `equals`/`hashCode`: `int[]` doesn't; use a `record Cell(int r, int c)`, `List.of(r, c)`, or encode `r * cols + c` (fastest).
- **Budget** — ≈ 10^8 simple operations/second after warm-up: n = 10^5 needs O(n log n); n ≤ 20 allows O(2^n).

Key facts & numbers

Pattern	Recognise it when	Complexity
Two pointers	sorted input; pair/triple with target sum; in-place O(1) space; palindrome; linked-list middle/cycle/k-th from end	O(n), O(1) space (+ sort)
Sliding window	longest/shortest contiguous substring/subarray "such that…"; at most k distinct; window of size k; **non-negative** sums	O(n)
Hashing	pairs in **unsorted** data; duplicates; frequencies; group by signature (anagrams); first unique; longest consecutive run	O(n) avg, O(n) space
Prefix sums	many range sums; "subarrays summing to k" (works with negatives: prefix + map of counts); 2D sums; difference array for range updates	O(n) build, O(1) query
Stack / monotonic stack	brackets; next greater/smaller; days until warmer; stock span; histogram; sliding-window max (monotonic deque)	O(n) — each index pushed/popped once
Heap / top-k / two heaps	k-th largest, k most frequent, k closest, merge k lists, Dijkstra; **two heaps**: running median (max-heap low half + min-heap high half), "best among currently eligible" (maximise capital)	top-k O(n log k) with a **min-heap of size k**; median O(log n) add, O(1) query
Binary search	sorted input; first/last position; rotated array; "minimum capacity/speed such that…" (search on the answer)	O(log n); on answer O(n log range)
Intervals	meetings, bookings, overlaps, min rooms, free time	O(n log n) sort + O(n) sweep
BFS / DFS / Dijkstra	levels, fewest steps (BFS); islands/regions, reachability (DFS); weighted cheapest route	O(V+E); Dijkstra O((V+E) log V)
Topological sort	prerequisites, dependencies, task order, cycle check, alien dictionary	O(V+E) (Kahn)
Union-find	connected components with edges arriving over time; "does this edge make a cycle?"; account merging; Kruskal	≈ O(1) per op (α(n)) with path compression + union by size
Dynamic programming	number of ways; min/max cost; longest subsequence; can you reach/make; branching brute force	states × work per state (coin change O(amount·coins), LCS O(m·n))
Backtracking	all subsets/permutations/combinations; N-Queens; Sudoku; word search; n ≤ ~20	subsets O(2ⁿ·n), permutations O(n!·n)
Tries	many prefix queries; autocomplete; word search II; shortest root replacement	O(L) per insert/lookup
Linked lists / fast & slow	k-th from end (gap of k + dummy node), intersection (walk A then B), palindrome list (reverse 2nd half, restore), cycle start (Floyd), happy number	O(n), O(1) space
Tree recursion	validate BST (pass `long` range down), LCA (split point bottom-up), max path sum (return one side, record both), serialise (preorder + `#`)	O(n), O(h) stack
Greedy	jump game (`farthest`), fewest jumps (range BFS), gas station (reset start when tank < 0), candies (two passes + `max`); prove it with an exchange argument, else DP	O(n) (+ sort)
Sort & search	k-th element without a full sort (quickselect, random pivot), sort 0/1/2 (Dutch flag), sort a linked list (merge sort)	quickselect O(n) avg / O(n²) worst; list merge sort O(n log n)

Pattern	Recognise it when	Complexity
Bit manipulation	appears twice except one (`^`), count 1-bits (`bits[i>>1] + (i&1)`), power of two (`n > 0 && (n & (n-1)) == 0`), swap bits (masks + `>>>`), subsets of n ≤ 20	O(1) per trick; subsets $O(2^n \cdot n)$
Math & geometry	spiral matrix (four boundaries), reverse int (check vs `MAX_VALUE / 10` before `* 10`), points on a line (gcd-reduced slope as a `long` key), Josephus (`J(n) = (J(n-1)+k) % n`)	spiral O(rows·cols); points $O(n^2)$

Java trap	Fix
`arr.length` vs `s.length()` vs `list.size()`	field, method, method
`(lo + hi) / 2` overflows	`lo + (hi - lo) / 2` or `(lo + hi) >>> 1`
`long x = n * n` (int × int overflows first)	`(long) n * n`
`Math.abs(Integer.MIN_VALUE)` is negative	work in `long` / `Math.absExact`
`Integer.MAX_VALUE` as ∞, then `+ 1`	guard, or use `MAX_VALUE / 2`
`-7 % 3 == -1`	`Math.floorMod`
`map.get(a) == map.get(b)` for 1000	`equals` (cache only −128…127)
`int x = map.get(missing)` → NPE	`getOrDefault`
`list.remove(1)` removes index	`remove(Integer.valueOf(1))`
Comparator `(a, b) -> a - b` / heap `(a, b) -> b - a`	`Integer.compare` , `comparingInt` , `reverseOrder()`
`split(".")`	`split("\\.")` ; `trim().split("\\s+")` for words
`'a' + 1` is `98`	cast: `(char) ('a' + 1)`
Deep recursion (≈10^4–10^5 frames)	explicit `ArrayDeque` stack (big grids)
`s += x` in a loop	`StringBuilder`

Code to remember

```java
int[] twoSum(int[] nums, int target) {                  // hashing
    Map<Integer, Integer> seen = new HashMap<>();
    for (int i = 0; i < nums.length; i++) {
        Integer j = seen.get(target - nums[i]);
        if (j != null) return new int[]{j, i};
        seen.put(nums[i], i);
    }
    return new int[0];
}
int kthLargest(int[] a, int k) {                        // min-heap of size k
    PriorityQueue<Integer> pq = new PriorityQueue<>();
    for (int x : a) { pq.offer(x); if (pq.size() > k) pq.poll(); }
    return pq.peek();
}
```

In the interview

Q: How do you talk through a problem? **A:** Clarify (sizes, ranges/overflow, duplicates, empty, sorted?) → work an example → brute force + its complexity → name the pattern → code with clear names → trace the example and edge cases (empty, one element, all equal, negatives, `MAX_VALUE`) → state time and space, including the sort and recursion stack.

Q: Why can `new PriorityQueue<Integer>((a, b) -> b - a)` give wrong answers? **A:** Subtraction overflows for values far apart; use `Comparator.reverseOrder()` .

Q: "Largest key ≤ x" while inserting? **A:** `TreeMap.floorKey(x)` / `TreeSet.floor(x)` , O(log n).

Q: Sliding window fails with negative numbers — what instead? **A:** Prefix sums with a `HashMap` of prefix counts.

Q: Why an explicit `ArrayDeque` for islands instead of recursion? **A:** A large grid can recurse ~10^6 deep → `StackOverflowError`.

Q: Streams in DSA code? **A:** Prefer plain loops: easier to trace, no boxing, early exits.

Q: When is greedy safe instead of DP? **A:** When an exchange argument holds — swapping the locally best choice into any optimal answer never makes it worse (jump game, gas station, intervals by end time). A small counter-example (coins {1, 3, 4}, amount 6) means DP.

Q: Quickselect or a heap for the k-th largest? **A:** Quickselect: O(n) average, O(n^2) worst, reorders the array. Size-k min-heap: O(n log k), works on a stream, never quadratic.

Q: Why does Floyd's second phase find the cycle start? **A:** At the meeting point slow has walked a + b and fast 2(a + b), so a + b is a multiple of the loop length; walking a more steps from the meeting point lands on the start, as does walking a from the head.

Gotchas

- Java polish: `import java.util.*`; declare interface types; return `new int[0]` / `List.of()`, not `null`; `var` only where obvious.
- Mutating the caller's input array → clone first.
- Relying on `HashMap` iteration order in an answer → `LinkedHashMap` / `TreeMap`.
- Removing inside for-each → `removeIf` / `Iterator.remove()`.
- Backtracking: add a **copy** of the path (`new ArrayList<>(path)`), un-choose after exploring.

TL;DR

Know the containers (`ArrayDeque`, `PriorityQueue`, `TreeMap`, `HashMap.merge`), the traps (overflow, `Integer ==`, `remove(int)`, subtraction comparators), and map problem wording to one of twenty patterns.

Last-minute checklist

- [ ] JDK ⊃ JRE ⊃ JVM; `javac` → bytecode (class version 69 = Java 25); interpreter → C1 → C2; LTS = 8, 11, 17, 21, 25.
- [ ] Eight primitives and `int` ≈ ±2.1 × 10^9; `/` truncates, `%` follows the dividend, overflow wraps silently (`Math.*Exact`, `long`).
- [ ] `0.1 + 0.2 != 0.3` → `BigDecimal` from strings, `compareTo`, `HALF_EVEN`.
- [ ] `==` is identity; `Integer` cache −128…127; unboxing `null` → NPE; `list.remove(int)` vs `remove(Object)`.
- [ ] Pass-by-value always; overloads chosen at compile time, overrides at run time.
- [ ] Immutable class recipe; `final` ≠ immutable; records are shallowly immutable.
- [ ] `equals` + `hashCode` together, same fields, immutable keys; `@Override` always.
- [ ] Interface vs abstract class; sealed + records + pattern `switch` = exhaustive dispatch.
- [ ] Arrow `switch` never falls through; switch expressions use `yield`; `case null` (21).
- [ ] Checked vs unchecked; try-with-resources (reverse close, suppressed); chain causes; never `return` from `finally`; restore the interrupt flag.
- [ ] Generics are invariant; PECS; erasure forbids `new T()`, `T[]`, `List<int>`.
- [ ] `HashMap`: 16 / 0.75 / treeify > 8 with table ≥ 64 / resize ×2; `ArrayList` grows ×1.5; `ArrayDeque` for stacks and queues; `PriorityQueue` is a min-heap with unsorted iteration.
- [ ] `List.of` (no nulls, unmodifiable) vs `Arrays.asList` (fixed-size view) vs `unmodifiableList` (view); sequenced collections (21).
- [ ] Streams: lazy, single use; `toMap` needs a merge function; `groupingBy` + downstream; `toList()` is unmodifiable; parallel only after measuring.

- [ ] `Optional` for return values only; `orElseGet` for expensive defaults.
- [ ] `synchronized` = exclusion + visibility; `volatile` = visibility only; `ConcurrentHashMap.merge`; bounded pools; `shutdown` vs `shutdownNow`.
- [ ] Happens-before; DCL needs `volatile` (or use the holder idiom); deadlock = lock order / `tryLock`.
- [ ] Virtual threads (21): I/O-bound, one per task, no pooling, `Semaphore` for limits; pinning fixed in 24; scoped values final in 25; structured concurrency still preview.
- [ ] Heap vs stack vs metaspace errors; G1 default, ZGC sub-ms; escape analysis; JMH for benchmarks; `jcmd` + JFR for diagnosis.
- [ ] Java 17 → 21 → 25 headline features, and which are still preview.
- [ ] Maven lifecycle and scopes; JUnit 5/6 lifecycle; mock vs stub vs spy; inject a `Clock`; `PreparedStatement` + pooled `DataSource`.
- [ ] Coding rounds: clarify → example → brute force → pattern → code → trace → complexity; watch overflow, `Integer ==`, subtraction comparators, deep recursion.